

True or false?

A box that is $\frac{1}{2}$ m by $\frac{1}{2}$ m by $\frac{1}{2}$ m has a volume of $\frac{1}{8}$ cubic meter.

A. True

B. False

SET 36 • 5.NF.B.6

FORWARD 2

Surface area of prisms with fractional edge lengths • Grade 7 • Geometry •
7.G.B.6

A rectangular prism is 4 cm by 3 cm by $2\frac{1}{2}$ cm.

What is its surface area?

Surface area: _____

A cube has edges that are $2\frac{1}{2}$ inches long.

What is its surface area?

Surface area: _____

A box is 5 in. by 4 in. by $1\frac{1}{2}$ in.

What is its surface area?

- A.** 30 square inches
- B.** 40 square inches
- C.** 67 square inches
- D.** $33\frac{1}{2}$ square inches

A cube has edges that are $1\frac{1}{2}$ m long.

What is its surface area?

Surface area: _____

True or false?

A prism that is 2 ft by $1\frac{1}{2}$ ft by 1 ft has a surface area of 13 square feet.

- A. True
- B. False

Divide a unit fraction by a whole number

MAIN	Grade 5 main question	5.NF.B.7.a
BACKWARD 1	Unit fractions get smaller as the number of parts grows	3.NF.A.1
BACKWARD 2	Sharing equally	3.OA.A.2
BACKWARD 4	Equivalent fractions	4.NF.A.1
FORWARD 1	Divide fractions	6.NS.A.1
FORWARD 2	Divide rational numbers	7.NS.A.2.b

SET 37 • 5.NF.B.7.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.a

Divide.

$$\frac{1}{7} \div 2$$

Quotient: _____

Divide.

$$\frac{1}{2} \div 5$$

Quotient: _____

$$\frac{1}{4} \div 2 = ?$$

A. $\frac{1}{2}$

B. $\frac{2}{4}$

C. $\frac{1}{8}$

D. 8

$\frac{1}{5}$ of a pan of brownies is shared equally by 3 friends.

What fraction of the whole pan does each friend get?

Fraction: _____

True or false?

$$\frac{1}{6} \div 3 = \frac{1}{2}$$

A. True

B. False

SET 37 • 5.NF.B.7.a

BACKWARD 1

Unit fractions get smaller as the number of parts grows • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

True or false?

When a whole is cut into more equal parts, each part is smaller.

A. True

B. False

Which unit fraction is the smallest?

A. $\frac{1}{6}$

B. $\frac{1}{8}$

C. $\frac{1}{2}$

D. $\frac{1}{3}$

A bar is cut into 6 equal parts.

What fraction of the bar is one part?

Fraction: _____

True or false?

$$\frac{1}{3} < \frac{1}{4}$$

A. True

B. False

What fraction of the bar is shaded?



Fraction: _____

SET 37 • 5.NF.B.7.a

BACKWARD 2

Sharing equally • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

15 is shared equally among 5 groups.

How much is in each group?

Answer: _____

True or false?

Sharing 8 equally between 2 groups gives 4 in each group.

A. True

B. False

Which expression shows 21 shared equally among 7 people?

A. $21 \div 7$

B. $21 - 7$

C. $7 \div 21$

D. 21×7

Divide.

$$16 \div 4$$

Quotient: _____

36 is shared equally among 6 people.

How much does each person get?

Answer: _____

SET 37 • 5.NF.B.7.a

BACKWARD 4

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Find the missing number.

$$\frac{1}{3} = \frac{?}{12}$$

Missing number: _____

True or false?

$$\frac{1}{2} = \frac{5}{10}$$

A. True

B. False

Which fraction is equal to $\frac{1}{4}$?

A. $\frac{4}{1}$

B. $\frac{1}{8}$

C. $\frac{2}{4}$

D. $\frac{2}{8}$

Find the missing number.

$$\frac{1}{5} = \frac{3}{?}$$

Missing number: _____

True or false?

$$\frac{1}{6} = \frac{3}{12}$$

A. True

B. False

SET 37 • 5.NF.B.7.a

FORWARD 1

Divide fractions • Grade 6 • The Number System • 6.NS.A.1

Divide.

$$\frac{3}{4} \div 3$$

Quotient: _____

Divide.

$$\frac{2}{5} \div 4$$

Quotient: _____

Divide.

$$\frac{5}{6} \div \frac{1}{3}$$

A. $\frac{5}{18}$

B. $2\frac{1}{2}$

C. $\frac{6}{9}$

D. $\frac{1}{2}$

$\frac{3}{4}$ pound of trail mix is shared equally among 6 bags.

How much trail mix is in each bag?

Pounds: _____

True or false?

$$\frac{2}{3} \div 2 = 1\frac{1}{3}$$

A. True

B. False

SET 37 • 5.NF.B.7.a

FORWARD 2

Divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Find the quotient.

$$-\frac{1}{3} \div 4$$

Quotient: _____

Find the quotient.

$$\frac{3}{5} \div \left(-\frac{3}{10}\right)$$

Quotient: _____

Find the quotient.

$$\left(-\frac{1}{2}\right) \div \left(-\frac{1}{4}\right)$$

A. $\frac{1}{8}$

B. $-\frac{1}{8}$

C. -2

D. 2

Find the quotient.

$$-\frac{4}{9} \div \frac{2}{3}$$

Quotient: _____

True or false?

$$-\left(\frac{1}{2} \div 5\right) = \left(-\frac{1}{2}\right) \div 5$$

A. True

B. False

Use models to divide a unit fraction by a whole number and check with multiplication

MAIN	Grade 5 main question	5.NF.B.7.a
BACKWARD 1	Unit fractions	3.NF.A.1
BACKWARD 2	A fraction as a multiple of a unit fraction	4.NF.B.4.a
FORWARD 1	Divide fractions and check with multiplication	6.NS.A.1
FORWARD 2	Divide rational numbers	7.NS.A.2.b

SET 69 • 5.NF.B.7.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.a

The bar shows $\frac{1}{3}$ shaded.

Split every third into 2 equal parts to find $\frac{1}{3} \div 2$. Then write the quotient.



Quotient: _____

Explain how you can use multiplication to check that

$$\frac{1}{4} \div 3 = \frac{1}{12}.$$

Explanation: _____

Which equation checks $\frac{1}{5} \div 4 = \frac{1}{20}$?

A. $\frac{1}{20} \div 4 = \frac{1}{80}$

B. $\frac{1}{5} + 4 = 4\frac{1}{5}$

C. $\frac{1}{20} \times 4 = \frac{1}{5}$

D. $\frac{1}{5} \times 4 = \frac{4}{5}$

True or false?

$\frac{1}{2} \div 3 = \frac{1}{6}$ because 3 groups of $\frac{1}{6}$ make $\frac{1}{2}$.

A. True

B. False

4 friends share $\frac{1}{2}$ of a pizza equally.

How much of the whole pizza does each friend get?
Describe a model that shows it.

Share: _____

Model: _____

SET 69 • 5.NF.B.7.a

BACKWARD 1

Unit fractions • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

A bar is cut into 8 equal parts.

What fraction of the bar is one part?

Fraction: _____

True or false?

2 parts of size $\frac{1}{6}$ make $\frac{2}{6}$.

A. True

B. False

What fraction of the bar is shaded?



A. $\frac{5}{6}$

B. $\frac{1}{5}$

C. $\frac{1}{6}$

D. $\frac{6}{1}$

A whole is cut into 3 equal parts.

What fraction of the whole is one part?

Fraction: _____

True or false?

$\frac{1}{4}$ is one of 3 equal parts of a whole.

A. True

B. False

SET 69 • 5.NF.B.7.a

BACKWARD 2

A fraction as a multiple of a unit fraction • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.a

Write $\frac{3}{12}$ as a whole number times $\frac{1}{12}$.

Expression: _____

True or false?

$$4 \times \frac{1}{8} = \frac{4}{8}$$

A. True

B. False

Which fraction equals $6 \times \frac{1}{10}$?

A. $\frac{1}{60}$

B. $\frac{6}{10}$

C. $\frac{10}{6}$

D. $\frac{6}{60}$

Multiply.

$$2 \times \frac{1}{6}$$

Product: _____

True or false?

$$3 \times \frac{1}{9} = \frac{3}{27}$$

A. True

B. False

SET 69 • 5.NF.B.7.a

FORWARD 1

Divide fractions and check with multiplication • Grade 6 • The Number System
• 6.NS.A.1

Divide.

$$\frac{2}{3} \div 4$$

Quotient: _____

True or false?

$$\frac{3}{4} \div \frac{1}{4} = 3 \text{ because } 3 \times \frac{1}{4} = \frac{3}{4}.$$

A. True

B. False

Which equation checks $\frac{2}{3} \div \frac{3}{4} = \frac{8}{9}$?

A. $\frac{8}{9} \times \frac{3}{4} = \frac{2}{3}$

B. $\frac{3}{4} \div \frac{8}{9} = \frac{27}{32}$

C. $\frac{8}{9} \times \frac{2}{3} = \frac{16}{27}$

D. $\frac{2}{3} \times \frac{3}{4} = \frac{1}{2}$

Divide.

$$\frac{4}{5} \div 2$$

Quotient: _____

True or false?

$$\frac{1}{2} \div \frac{1}{3} = \frac{1}{6}$$

A. True

B. False

SET 69 • 5.NF.B.7.a

FORWARD 2

Divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Find the quotient.

$$-\frac{1}{4} \div 2$$

Quotient: _____

True or false?

$$-\frac{1}{3} \div 5 = -\frac{1}{15}$$

A. True

B. False

Find the quotient.

$$\frac{1}{6} \div (-3)$$

A. $\frac{1}{18}$

B. $-\frac{1}{2}$

C. -18

D. $-\frac{1}{18}$

Find the quotient.

$$\left(-\frac{1}{2}\right) \div (-4)$$

Quotient: _____

True or false?

$$\left(-\frac{1}{5}\right) \div 2 = \frac{1}{10}$$

A. True

B. False

Divide a whole number by a unit fraction

MAIN

Grade 5 main question

5.NF.B.7.b

BACKWARD 1

Division as "how many groups?"

3.OA.A.2

BACKWARD 2

Unit fractions that make a whole

3.NF.A.1

FORWARD 1

Divide whole numbers by fractions

6.NS.A.1

FORWARD 2

Unit rates from quantities measured in fractions

7.RP.A.1

SET 38 • 5.NF.B.7.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.b

Divide.

$$6 \div \frac{1}{5}$$

Quotient: _____

Divide.

$$3 \div \frac{1}{6}$$

Quotient: _____

How many $\frac{1}{3}$ s are in 7?

A. 10

B. 21

C. $\frac{7}{3}$

D. $\frac{1}{21}$

Divide.

$$9 \div \frac{1}{4}$$

Quotient: _____

True or false?

$$5 \div \frac{1}{2} = 2\frac{1}{2}$$

A. True

B. False

SET 38 • 5.NF.B.7.b

BACKWARD 1

Division as "how many groups?" • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.2

How many groups of 3 are in 18?

Groups: _____

True or false?

$20 \div 5$ can mean "How many 5s are in 20?"

A. True

B. False

Pam puts 24 eggs into cartons of 6.

Which equation finds the number of cartons?

A. $6 \div 24 = \square$

B. $24 - 6 = \square$

C. $24 \times 6 = \square$

D. $24 \div 6 = \square$

How many 4s are in 32?

Answer: _____

A 30-inch ribbon is cut into 5-inch pieces.
How many pieces are there?

Pieces: _____

SET 38 • 5.NF.B.7.b

BACKWARD 2

Unit fractions that make a whole • Grade 3 • Number & Operations–Fractions •
3.NF.A.1

How many $\frac{1}{4}$ s make 1 whole?

Answer: _____

True or false?

There are 3 thirds in 2 wholes.

A. True

B. False

How many $\frac{1}{6}$ s make 2 wholes?

- A.** 2
- B.** 8
- C.** 12
- D.** 6

How many $\frac{1}{8}$ s make 1 whole?

Answer: _____

True or false?

There are 2 halves in 2 wholes.

A. True

B. False

SET 38 • 5.NF.B.7.b

FORWARD 1

Divide whole numbers by fractions • Grade 6 • The Number System • 6.NS.A.1

Divide.

$$6 \div \frac{3}{4}$$

Quotient: _____

Divide.

$$4 \div \frac{2}{5}$$

Quotient: _____

How many $\frac{2}{3}$ -cup servings are in 8 cups?

A. 6

B. $\frac{1}{12}$

C. $\frac{16}{3}$

D. 12

Divide.

$$5 \div \frac{5}{6}$$

Quotient: _____

True or false?

$$3 \div \frac{3}{8} = \frac{9}{8}$$

A. True

B. False

SET 38 • 5.NF.B.7.b

FORWARD 2

Unit rates from quantities measured in fractions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A car uses $\frac{1}{4}$ gallon of gas to go 6 miles.

How many miles does it go per gallon?

Miles per gallon: _____

A machine makes 5 parts every $\frac{1}{3}$ minute.

How many parts does it make per minute?

Parts per minute: _____

Lee runs 2 miles in $\frac{1}{6}$ hour.

What is his speed?

- A.** $\frac{1}{12}$ mile per hour
- B.** 3 miles per hour
- C.** 8 miles per hour
- D.** 12 miles per hour

A hose fills 10 liters in $\frac{1}{2}$ minute.

How many liters does it fill per minute?

Liters per minute: _____

True or false?

Walking 3 miles in $\frac{3}{4}$ hour is a speed of 4 miles per hour.

- A. True
- B. False

Use models to divide a whole number by a unit fraction and check with multiplication

MAIN	Grade 5 main question	5.NF.B.7.b
BACKWARD 1	Write whole numbers as sums of unit fractions	4.NF.B.3.b
BACKWARD 2	A fraction as a multiple of a unit fraction	4.NF.B.4.a
FORWARD 1	Divide whole numbers by fractions and check with multiplication	6.NS.A.1
FORWARD 2	Divide rational numbers	7.NS.A.2.b

SET 70 • 5.NF.B.7.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.b

Each bar is 1 whole.

Split each whole into fourths to find $2 \div \frac{1}{4}$. Then write the quotient.

1 whole



1 whole



Quotient: _____

Explain how to use multiplication to check that $5 \div \frac{1}{3} = 15$.

Explanation: _____

Which equation checks $6 \div \frac{1}{2} = 12$?

A. $12 \div \frac{1}{2} = 24$

B. $6 \times \frac{1}{2} = 3$

C. $12 \times \frac{1}{2} = 6$

D. $6 \times 12 = 72$

True or false?

$3 \div \frac{1}{5} = 15$ because each of the 3 wholes has 5 fifths.

A. True

B. False

Write a story problem for $4 \div \frac{1}{2}$ and solve it. Describe a model that shows the answer.

Story and answer: _____

Model: _____

SET 70 • 5.NF.B.7.b

BACKWARD 1

Write whole numbers as sums of unit fractions • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.b

Write 1 as a sum of thirds.

Sum: _____

True or false?

$$2 = \frac{5}{5} + \frac{5}{5}$$

A. True

B. False

Which sum equals 1?

A. $\frac{1}{4} + \frac{1}{4} + \frac{1}{4}$

B. $\frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4}$

C. $\frac{1}{4} + \frac{1}{4}$

D. $\frac{1}{2} + \frac{1}{4}$

Write 2 as a fraction with a denominator of 6.

Fraction: _____

True or false?

$$3 = \frac{3}{3}$$

A. True

B. False

SET 70 • 5.NF.B.7.b

BACKWARD 2

A fraction as a multiple of a unit fraction • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.a

Write $\frac{8}{4}$ as a whole number times $\frac{1}{4}$.

Expression: _____

True or false?

$$15 \times \frac{1}{3} = 5$$

A. True

B. False

What is $10 \times \frac{1}{5}$?

A. 2

B. $\frac{1}{50}$

C. 50

D. $\frac{10}{50}$

Multiply.

$$6 \times \frac{1}{2}$$

Product: _____

True or false?

$$12 \times \frac{1}{4} = 48$$

A. True

B. False

SET 70 • 5.NF.B.7.b

FORWARD 1

Divide whole numbers by fractions and check with multiplication • Grade 6 •
The Number System • 6.NS.A.1

Divide.

$$3 \div \frac{3}{5}$$

Quotient: _____

True or false?

$2 \div \frac{2}{3} = 3$ because $3 \times \frac{2}{3} = 2$.

A. True

B. False

Which equation checks $4 \div \frac{2}{7} = 14$?

A. $4 \times 14 = 56$

B. $4 \times \frac{2}{7} = \frac{8}{7}$

C. $14 \div \frac{2}{7} = 49$

D. $14 \times \frac{2}{7} = 4$

How many $\frac{3}{4}$ -foot pieces can be cut from a 9-foot board?

Pieces: _____

True or false?

$$6 \div \frac{3}{4} = \frac{9}{2}$$

A. True

B. False

SET 70 • 5.NF.B.7.b

FORWARD 2

Divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Find the quotient.

$$-3 \div \frac{1}{4}$$

Quotient: _____

True or false?

$$4 \div \left(-\frac{1}{2}\right) = -8$$

A. True

B. False

Find the quotient.

$$(-5) \div \left(-\frac{1}{3}\right)$$

A. 15

B. $\frac{5}{3}$

C. -15

D. $-\frac{5}{3}$

Find the quotient.

$$-2 \div \frac{1}{6}$$

Quotient: _____

True or false?

$$(-6) \div \frac{1}{2} = -3$$

A. True

B. False

Solve word problems that divide unit fractions and whole numbers

MAIN	Grade 5 main question	5.NF.B.7.c
BACKWARD 1	Division word problems	3.OA.A.3
BACKWARD 3	Word problems: a whole number times a unit fraction	4.NF.B.4.c
FORWARD 1	Word problems that divide fractions by fractions	6.NS.A.1
FORWARD 2	Real-world problems that divide rational numbers	7.NS.A.3

SET 39 • 5.NF.B.7.c

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.c

How many $\frac{1}{4}$ -cup scoops of birdseed are in 3 cups?

Scoops: _____

4 students share $\frac{1}{3}$ liter of paint equally.

How much paint does each student get?

Liters: _____

A 5-meter rope is cut into pieces that are $\frac{1}{4}$ meter long.

How many pieces are there?

A. 20

B. $\frac{1}{20}$

C. $\frac{5}{4}$

D. 9

$\frac{1}{8}$ of a pizza is shared equally by 2 people.

What fraction of the whole pizza does each person get?

Fraction: _____

True or false?

A 6-mile trail is split into $\frac{1}{2}$ -mile sections. There are 12 sections.

- A.** True
- B.** False

SET 39 • 5.NF.B.7.c

BACKWARD 1

Division word problems • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

Kim has 28 beads. Each bracelet uses 7 beads.
How many bracelets can she make?

Bracelets: _____

True or false?

If 32 students sit 4 at each table, there are 6 tables.

A. True

B. False

Raj shares 40 cards equally among 5 friends.

How many cards does each friend get?

- A.** 8
- B.** 200
- C.** 35
- D.** 45

A 27-foot fence is split into 9 equal sections.
How long is each section?

Length: _____

48 eggs are packed 6 to a carton.

How many cartons are filled?

Cartons: _____

SET 39 • 5.NF.B.7.c

BACKWARD 3

Word problems: a whole number times a unit fraction • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.c

Each scoop holds $\frac{1}{4}$ cup.

How many cups are in 8 scoops?

Cups: _____

True or false?

5 servings of $\frac{1}{3}$ pound each weigh $\frac{5}{3}$ pounds in all.

A. True

B. False

Each piece of ribbon is $\frac{1}{2}$ meter long.

How long are 6 pieces in all?

A. $\frac{1}{12}$ meter

B. $6\frac{1}{2}$ meters

C. 12 meters

D. 3 meters

Each lap is $\frac{1}{8}$ mile.

How far are 16 laps?

Distance: _____

True or false?

4 cups that each hold $\frac{1}{3}$ liter hold $\frac{4}{12}$ liter in all.

A. True

B. False

SET 39 • 5.NF.B.7.c

FORWARD 1

Word problems that divide fractions by fractions • Grade 6 • The Number System • 6.NS.A.1

How many $\frac{3}{8}$ -pound bags can be filled from 3 pounds of rice?

Bags: _____

$\frac{2}{3}$ of a cake is shared equally by 4 people.

What fraction of the whole cake does each person get?

Fraction: _____

A path is $\frac{9}{10}$ mile long. It is split into sections that are $\frac{3}{10}$ mile long.

How many sections are there?

A. $\frac{27}{100}$

B. 6

C. $\frac{1}{3}$

D. 3

A rectangle has an area of 2 square yards and a length of $2\frac{1}{2}$ yards.

What is its width?

Width: _____

True or false?

$\frac{3}{4}$ cup of oil fills exactly 3 bottles that hold $\frac{1}{4}$ cup each.

A. True

B. False

SET 39 • 5.NF.B.7.c

FORWARD 2

Real-world problems that divide rational numbers • Grade 7 • The Number System • 7.NS.A.3

A diver descends 18 feet in $\frac{1}{2}$ minute at a constant rate.

What is the change in depth per minute?

Change per minute: _____

The temperature fell 5°F in $\frac{1}{4}$ hour.

At this rate, what is the change in temperature per hour?

Change per hour: _____

A bank account changes by $-\$45$ in $\frac{1}{3}$ month at a steady rate.

What is the change per month?

A. $\$135$

B. $-\$45$

C. $-\$15$

D. $-\$135$

Find the value.

$$-6 \div \frac{1}{3}$$

Value: _____

True or false?

$$-2 \div \frac{1}{4} = -\frac{1}{2}$$

A. True

B. False

Convert from a larger unit to a smaller unit

MAIN	Grade 5 main question	5.MD.A.1
BACKWARD 1	Relative sizes of measurement units	4.MD.A.1
BACKWARD 2	Multiplying whole numbers	4.NBT.B.5
BACKWARD 4	Multiply whole numbers by 10, 100, and 1,000	4.NBT.A.1
FORWARD 1	Use ratio reasoning to convert measurement units	6.RP.A.3.d
FORWARD 2	Unit rates with fractional quantities and different units	7.RP.A.1

SET 40 • 5.MD.A.1

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.A.1

How many inches are in $4\frac{1}{2}$ feet?

Inches: _____

How many ounces are in 3 pounds?

Ounces: _____

A track is 2.5 kilometers long.

How many meters is that?

- A.** 250 m
- B.** 25,000 m
- C.** 2,500 m
- D.** 25 m

Convert.

6 yards = ____ feet

Feet: _____

A jug holds 3 gallons.

How many quarts is that?

Quarts: _____

SET 40 • 5.MD.A.1

BACKWARD 1

Relative sizes of measurement units • Grade 4 • Measurement & Data •
4.MD.A.1

How many inches are in 1 foot?

Inches: _____

True or false?

1 pound = 16 ounces

A. True

B. False

How many quarts are in 1 gallon?

- A.** 2
- B.** 16
- C.** 8
- D.** 4

How many meters are in 1 kilometer?

Meters: _____

True or false?

1 yard = 12 inches

A. True

B. False

SET 40 • 5.MD.A.1

BACKWARD 2

Multiplying whole numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

Multiply.

$$16 \times 3$$

Product: _____

Multiply.

$$1,000 \times 7$$

Product: _____

True or false?

$$3 \times 36 = 98$$

A. True

B. False

$$12 \times 9 = ?$$

- A.** 96
- B.** 118
- C.** 21
- D.** 108

Multiply.

$$60 \times 5$$

Product: _____

SET 40 • 5.MD.A.1

BACKWARD 4

Multiply whole numbers by 10, 100, and 1,000 • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1

Multiply.

$$25 \times 100$$

Product: _____

True or false?

$$7 \times 1,000 = 700$$

A. True

B. False

$$42 \times 10 = ?$$

- A.** 4.2
- B.** 420
- C.** 4,200
- D.** 42

Multiply.

$$16 \times 1,000$$

Product: _____

True or false?

3×100 is 10 times as much as 3×10 .

A. True

B. False

SET 40 • 5.MD.A.1

FORWARD 1

Use ratio reasoning to convert measurement units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

There are 2.54 centimeters in 1 inch.

How many centimeters are in 5 inches?

Centimeters: _____

A car travels 3 miles every 4 minutes.

At this rate, how many miles does it travel in 1 hour?

Miles: _____

There are 3 feet in 1 yard.

How many yards are in 27 feet?

- A.** 81
- B.** 30
- C.** 9
- D.** 24

1 kilogram is about 2.2 pounds.

About how many pounds are in 15 kilograms?

Pounds: _____

True or false?

If 1 cup = 8 fluid ounces, then 5 cups = 13 fluid ounces.

- A. True
- B. False

SET 40 • 5.MD.A.1

FORWARD 2

Unit rates with fractional quantities and different units • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A snail moves 6 feet in $\frac{1}{2}$ hour.

What is its speed in feet per minute?

Speed: _____

A cyclist rides $\frac{3}{4}$ mile in 3 minutes.

What is the speed in miles per hour?

Speed: _____

A faucet drips $\frac{1}{2}$ cup every 5 minutes.

How many cups does it drip per hour?

A. $2\frac{1}{2}$

B. $\frac{1}{10}$

C. 6

D. 10

A plant grows $\frac{1}{4}$ inch per week.

How many inches does it grow in 52 weeks?

Inches: _____

True or false?

Running $1\frac{1}{2}$ miles in 15 minutes is a speed of 6 miles per hour.

- A. True
- B. False

Convert from a smaller unit to a larger unit

MAIN	Grade 5 main question	5.MD.A.1
BACKWARD 1	Relative sizes of measurement units	4.MD.A.1
BACKWARD 2	Dividing by a one-digit number	4.NBT.B.6
BACKWARD 4	Write tenths and hundredths as decimals	4.NF.C.6
FORWARD 1	Use ratio reasoning to convert measurement units	6.RP.A.3.d
FORWARD 2	Equations for unit conversions	7.RP.A.2.c

SET 41 • 5.MD.A.1

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.A.1

Convert.

30 inches = ____ feet

Feet: _____

Convert.

750 milliliters = ____ liters

Liters: _____

A rope is 45 feet long.

How many yards is that?

A. 15

B. 5

C. 135

D. 42

Convert.

40 ounces = ____ pounds

Pounds: _____

Convert.

1,350 grams = ____ kilograms

Kilograms: _____

SET 41 • 5.MD.A.1

BACKWARD 1

Relative sizes of measurement units • Grade 4 • Measurement & Data •
4.MD.A.1

How many milliliters are in 1 liter?

Milliliters: _____

True or false?

1 kilogram = 1,000 grams

A. True

B. False

How many feet are in 1 yard?

A. 10

B. 3

C. 36

D. 12

How many cups are in 1 pint?

Cups: _____

True or false?

1 foot = 10 inches

A. True

B. False

SET 41 • 5.MD.A.1

BACKWARD 2

Dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$36 \div 3$$

Quotient: _____

Divide.

$$64 \div 4$$

Quotient: _____

True or false?

$$84 \div 7 = 14$$

A. True

B. False

$$90 \div 3 = ?$$

- A.** 33
- B.** 300
- C.** 27
- D.** 30

Divide.

$$5,000 \div 5$$

Quotient: _____

SET 41 • 5.MD.A.1

BACKWARD 4

Write tenths and hundredths as decimals • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{75}{100}$ as a decimal.

Decimal: _____

True or false?

$$\frac{5}{10} = 0.5$$

A. True

B. False

Which decimal is equal to $\frac{45}{100}$?

- A.** 45
- B.** 0.045
- C.** 0.45
- D.** 4.5

Write $2\frac{6}{10}$ as a decimal.

Decimal: _____

True or false?

$$\frac{8}{10} = 0.08$$

A. True

B. False

SET 41 • 5.MD.A.1

FORWARD 1

Use ratio reasoning to convert measurement units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

There are 12 inches in 1 foot.

How many feet are in 102 inches?

Feet: _____

There are 1,000 meters in 1 kilometer.

How many kilometers are in 4,200 meters?

Kilometers: _____

A recipe needs 10 cups of milk. There are 4 cups in 1 quart.

How many quarts of milk are needed?

A. 40

B. 6

C. $2\frac{1}{2}$

D. 14

A trail is 5 kilometers long. 1 kilometer is about 0.62 mile.

About how many miles long is the trail?

Miles: _____

True or false?

There are 60 minutes in 1 hour, so 150 minutes = $1\frac{1}{2}$ hours.

A. True

B. False

SET 41 • 5.MD.A.1

FORWARD 2

Equations for unit conversions • Grade 7 • Ratios & Proportional Relationships •
7.RP.A.2.c

There are 2.54 centimeters in 1 inch.

Write an equation for the number of centimeters c in i inches.

Equation: _____

Write an equation for the number of feet f in y yards.

Equation: _____

Which equation converts k kilograms to g grams?

A. $k = 1,000g$

B. $g = 1,000k$

C. $g = k \div 1,000$

D. $g = k + 1,000$

Write an equation for the number of pounds p in z ounces.

Equation: _____

True or false?

The equation $m = 60h$ gives the number of minutes m in h hours.

- A. True
- B. False

Solve multistep problems that involve converting units

MAIN	Grade 5 main question	5.MD.A.1
BACKWARD 1	Word problems with measurement	4.MD.A.2
BACKWARD 2	Relative sizes of measurement units	4.MD.A.1
FORWARD 1	Convert units to compare rates	6.RP.A.3.d
FORWARD 2	Multistep ratio problems	7.RP.A.3

SET 42 • 5.MD.A.1

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.A.1

Ana has a 3-yard ribbon. She cuts off 4 feet.
How many feet of ribbon are left?

Feet: _____

A recipe needs 2 pounds of flour. Jo has 20 ounces.
How many more ounces does she need?

Ounces: _____

A runner ran 1.2 km and then 850 m.

How many meters did she run in all?

- A.** 1,050 m
- B.** 2,050 m
- C.** 9,700 m
- D.** 862 m

A jug holds 2 liters of water. Six glasses of 250 milliliters each are poured from it.

How many milliliters are left in the jug?

Milliliters: _____

A board is 8 feet long. It is cut into pieces that are 16 inches long.

How many pieces are there?

Pieces: _____

SET 42 • 5.MD.A.1

BACKWARD 1

Word problems with measurement • Grade 4 • Measurement & Data •
4.MD.A.2

A movie starts at 2:15 and lasts 1 hour 40 minutes.

What time does it end?

Time: _____

True or false?

3 quarters and 2 dimes make 85 cents.

A. True

B. False

Jake has 2 meters of string. He uses 75 cm.

How many centimeters are left?

A. 1.25 cm

B. 75 cm

C. 125 cm

D. 275 cm

A bag holds 3 kilograms of apples.

How many grams is that?

Grams: _____

Liz runs 400 meters four times.

How many meters does she run in all?

Meters: _____

SET 42 • 5.MD.A.1

BACKWARD 2

Relative sizes of measurement units • Grade 4 • Measurement & Data •
4.MD.A.1

How many ounces are in 2 pounds?

Ounces: _____

True or false?

1 gallon = 4 quarts

A. True

B. False

How many centimeters are in 1 meter?

- A.** 1,000
- B.** 100
- C.** 10
- D.** 12

How many feet are in 3 yards?

Feet: _____

True or false?

1 liter = 100 milliliters

A. True

B. False

SET 42 • 5.MD.A.1

FORWARD 1

Convert units to compare rates • Grade 6 • Ratios & Proportional Relationships
• 6.RP.A.3.d

A car uses 2 gallons of gas to drive 64 miles. There are 4 quarts in 1 gallon.

How many miles does the car drive per quart?

Miles: _____

Mo walks 3 kilometers in 40 minutes.

At this rate, how many meters does he walk per minute?

Meters per minute: _____

Fabric costs \$6 per yard.

How much do 12 feet of fabric cost?

A. \$72

B. \$18

C. \$24

D. \$2

A 24-ounce bag of rice costs \$3. There are 16 ounces in 1 pound.

What is the cost per pound?

Cost per pound: _____

True or false?

60 miles per hour is the same speed as 1 mile per minute.

- A. True
- B. False

SET 42 • 5.MD.A.1

FORWARD 2

Multistep ratio problems • Grade 7 • Ratios & Proportional Relationships •
7.RP.A.3

A 2-liter bottle of soda costs \$1.80. A 500-milliliter can costs \$0.60.

Which is the better buy per liter?

Better buy: _____

A map scale is 1 cm : 2 km. Two towns are 7.5 cm apart on the map. A car drives 60 km per hour.

How many minutes does the drive take?

Minutes: _____

One gallon of paint covers 350 square feet.

How many gallons are needed to cover 1,400 square feet?

A. 1,050

B. 3

C. 4

D. 5

A recipe for 4 people uses 600 grams of pasta.
How many kilograms are needed for 10 people?

Kilograms: _____

True or false?

At 5 miles per hour, a 2.5-mile walk takes 50 minutes.

A. True

B. False

Make a line plot of measurements in fractions of a unit

MAIN	Grade 5 main question	5.MD.B.2
BACKWARD 1	Reading a line plot with fractions	4.MD.B.4
BACKWARD 2	Fractions on a number line	3.NF.A.2
BACKWARD 3	Line plots of whole-number measurements	2.MD.D.9
FORWARD 1	Display numerical data in a dot plot	6.SP.B.4
FORWARD 2	Compare the difference in centers with the variability	7.SP.B.3

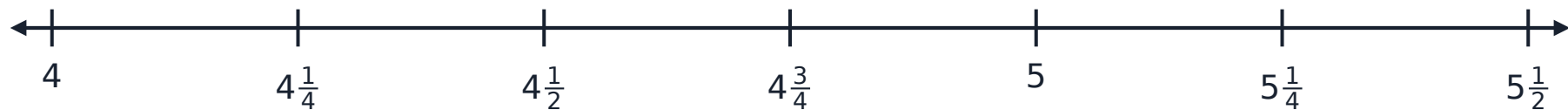
SET 43 • 5.MD.B.2

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.B.2

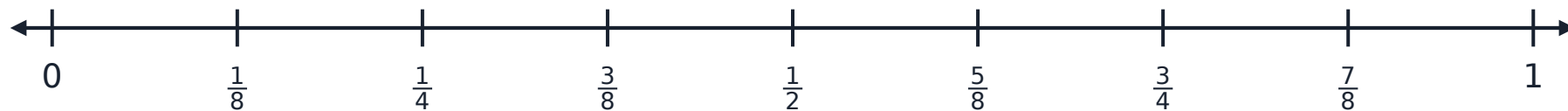
Pencil lengths (inches): $4\frac{1}{2}$, 5, $4\frac{1}{2}$, $5\frac{1}{4}$, $4\frac{3}{4}$, 5, $4\frac{1}{2}$

Make a line plot of the data.



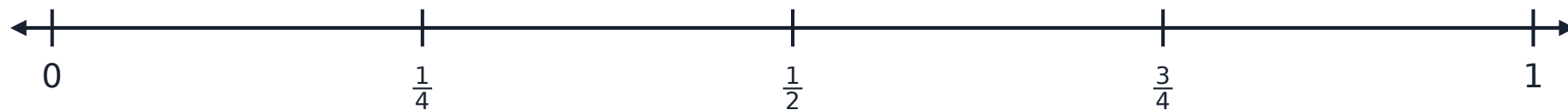
Rainfall (inches): $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{8}$, $\frac{1}{4}$, $\frac{3}{8}$, $\frac{1}{4}$, $\frac{1}{2}$

Make a line plot of the data.



Apple weights (pounds): $\frac{3}{4}$, $\frac{1}{2}$, $\frac{3}{4}$, 1 , $\frac{1}{2}$, $\frac{3}{4}$, $\frac{1}{4}$

Make a line plot of the data.



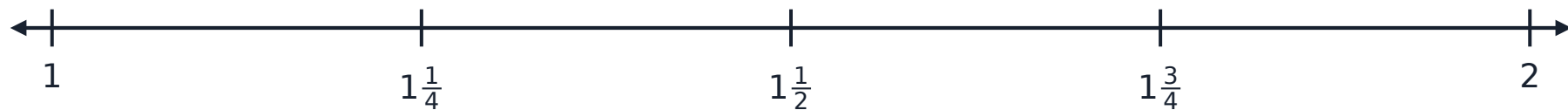
Lengths (inches): $\frac{1}{8}$, $\frac{3}{8}$, $\frac{3}{8}$, $\frac{5}{8}$, $\frac{3}{8}$

In a line plot, how many X's are drawn at $\frac{3}{8}$?

X's: _____

Water in each bottle (liters): $1\frac{1}{2}$, 2, $1\frac{1}{4}$, $1\frac{1}{2}$, $1\frac{3}{4}$, 2, $1\frac{1}{2}$

Make a line plot of the data.



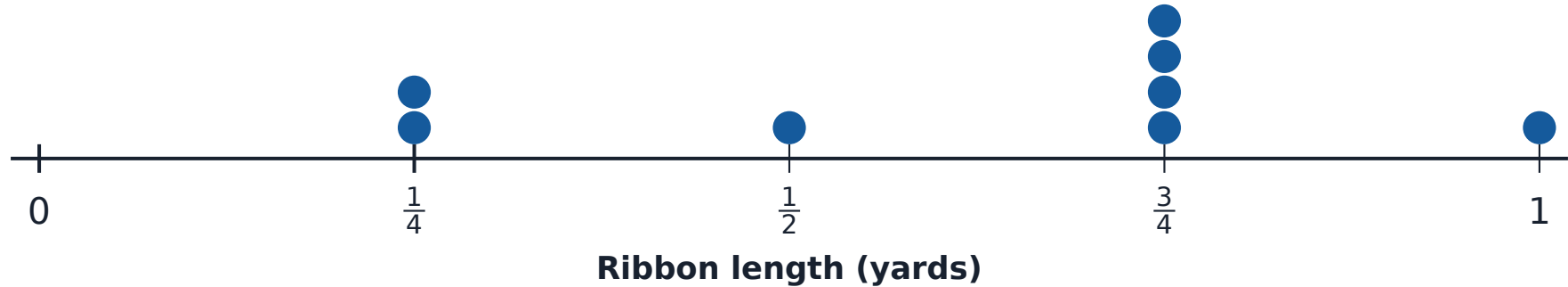
SET 43 • 5.MD.B.2

BACKWARD 1

Reading a line plot with fractions • Grade 4 • Measurement & Data • 4.MD.B.4

The line plot shows ribbon lengths.

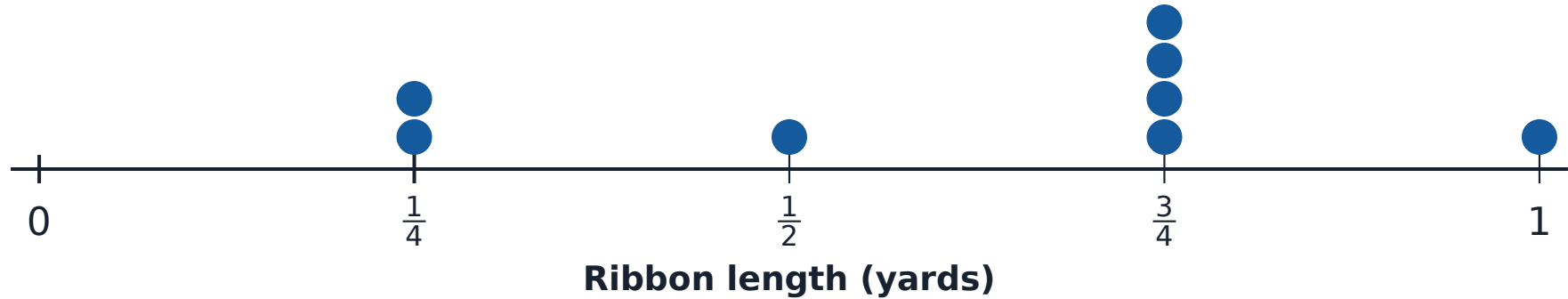
How many ribbons are $\frac{3}{4}$ yard long?



Ribbons: _____

True or false?

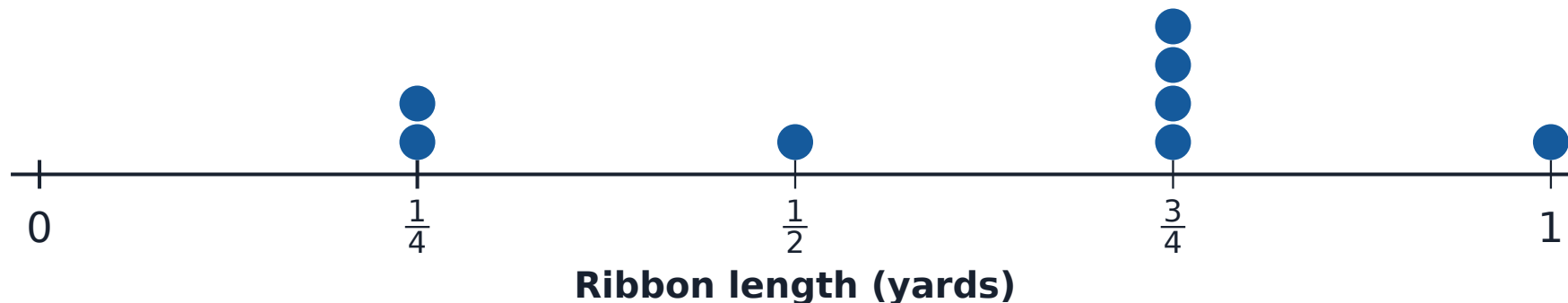
Two ribbons are $\frac{1}{4}$ yard long.



A. True

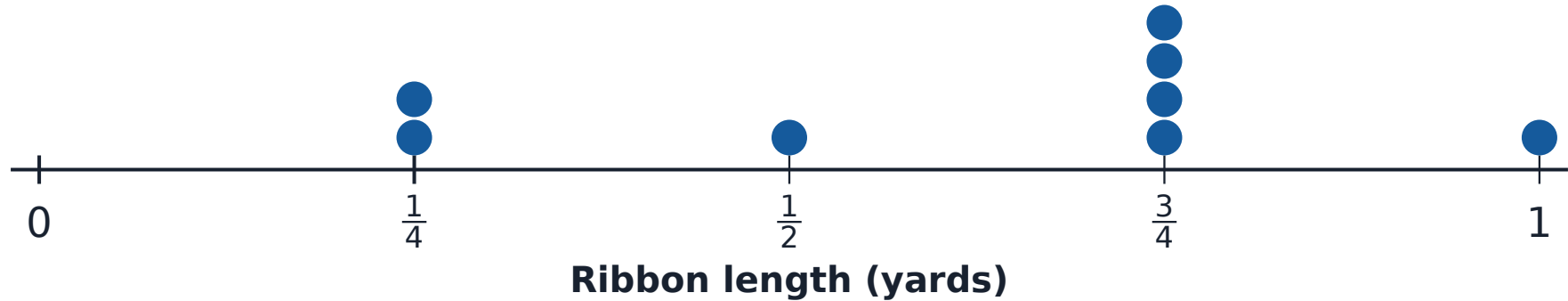
B. False

Which ribbon length is the most common?



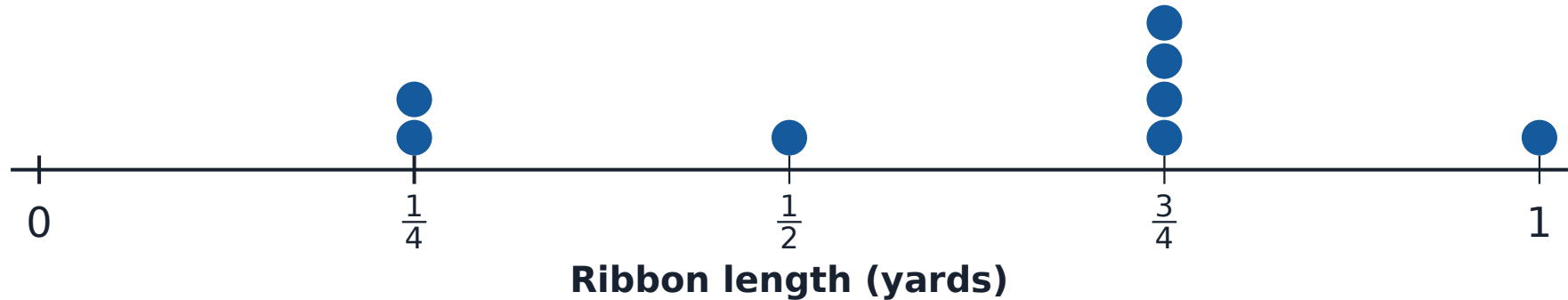
- A.** 1 yard **B.** $\frac{1}{2}$ yard **C.** $\frac{1}{4}$ yard **D.** $\frac{3}{4}$ yard

How many ribbons were measured?



Ribbons: _____

What is the difference between the longest and the shortest ribbon?



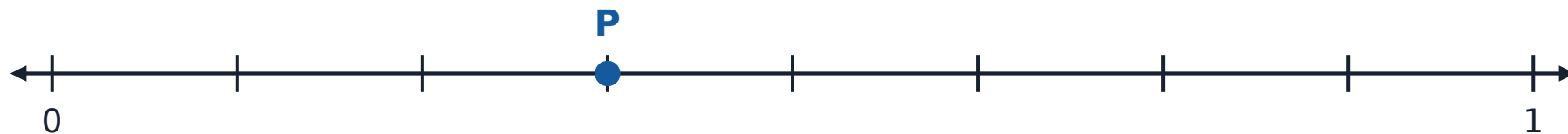
Difference: _____

SET 43 • 5.MD.B.2

BACKWARD 2

Fractions on a number line • Grade 3 • Number & Operations–Fractions •
3.NF.A.2

What fraction is at point P?



P = _____

True or false?

$\frac{2}{4}$ and $\frac{1}{2}$ are at the same point on a number line.

A. True

B. False

Into how many equal parts is the space from 0 to 1 divided?



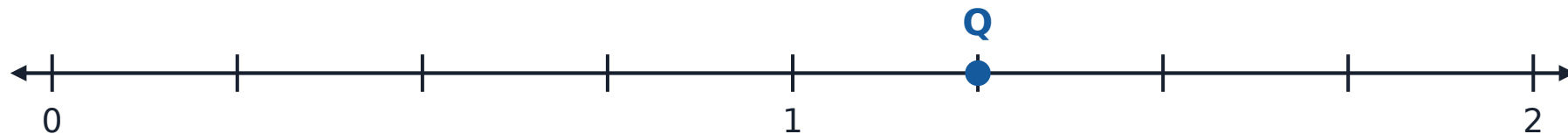
A. 5

B. 7

C. 6

D. 1

What number is at point Q?



Q = _____

True or false?

On a number line from 0 to 1 marked in eighths, $\frac{6}{8}$ is one mark to the left of 1.

- A.** True
- B.** False

SET 43 • 5.MD.B.2

BACKWARD 3

Line plots of whole-number measurements • Grade 2 • Measurement & Data •
2.MD.D.9

Lengths (inches): 5, 6, 6, 8, 6

In a line plot, how many X's are drawn at 6?

X's: _____

True or false?

A line plot shows each measurement as a mark above a number line.

- A.** True
- B.** False

Which number line would you use for a line plot of 3, 4, 4, 7?

- A.** A number line from 0 to 8
- B.** A number line from 0 to 3
- C.** A number line from 5 to 6
- D.** A number line from 10 to 20

Lengths (cm): 2, 4, 4, 5

Make a line plot of the data.



Lengths (feet): 9, 10, 10, 10, 12

Which length gets the most X's?

Length: _____

SET 43 • 5.MD.B.2

FORWARD 1

Display numerical data in a dot plot • Grade 6 • Statistics & Probability •
6.SP.B.4

Hours of sleep: 8, 9, 9, 10, 8, 9, 7

Make a dot plot of the data.



Number of siblings: 0, 1, 1, 2, 3, 1, 0, 2

Make a dot plot of the data.



Which number line fits a dot plot of 12, 15, 15, 18, 20?

- A.** A number line from 0 to 5
- B.** A number line from 20 to 40
- C.** A number line from 10 to 20
- D.** A number line from 15 to 18

Data: 4, 6, 6, 6, 7, 9

In a dot plot, how many dots go at 6?

Dots: _____

True or false?

A dot plot shows every value in a data set.

- A. True
- B. False

SET 43 • 5.MD.B.2

FORWARD 2

Compare the difference in centers with the variability • Grade 7 • Statistics & Probability • 7.SP.B.3

Plot A has a mean of 5 inches and Plot B has a mean of 8 inches. Each plot has a mean absolute deviation (MAD) of 1 inch.

The difference in the means is how many times the MAD?



Plot A: plant heights (inches)



Plot B: plant heights (inches)

Times the MAD: _____

True or false?

The difference in the means (3 inches) is 3 times the MAD, so the two distributions overlap only a little.



Plot A: plant heights (inches)



Plot B: plant heights (inches)

A. True

B. False

Which statement compares the two plots correctly?



Plot A: plant heights (inches)



Plot B: plant heights (inches)

- A.** The centers differ by less than 1 MAD, so the plots are about the same.
- B.** Plot A has much more variation than Plot B.
- C.** The plots cannot be compared because they show different plants.
- D.** The centers differ by 3 times the MAD, so Plot B plants are noticeably taller.

Two teams have mean heights of 150 cm and 156 cm.
Each team has a MAD of 3 cm.

The difference in the means is how many times the
MAD?

Times the MAD: _____

True or false?

Two data sets have means of 20 and 21, and each has a MAD of 4. The difference in the means is large compared with the variability.

- A.** True
- B.** False

Use operations on fractions to solve problems with line plot data

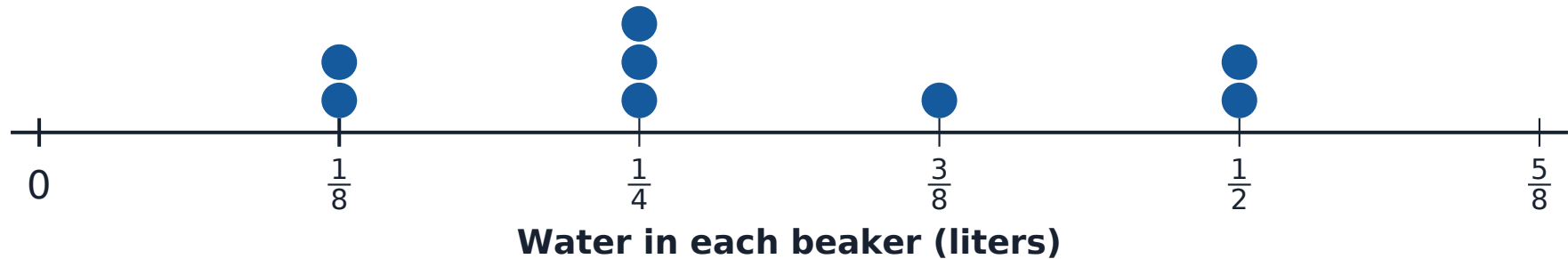
MAIN	Grade 5 main question	5.MD.B.2
BACKWARD 1	Solving problems from a line plot with like denominators	4.MD.B.4
BACKWARD 4	Rename halves and fourths as eighths	4.NF.A.1
BACKWARD 5	Add fractions with like denominators	4.NF.B.3.a
FORWARD 1	Find the mean of a data set	6.SP.B.5.c
FORWARD 2	Compare two populations using means	7.SP.B.4

SET 44 • 5.MD.B.2

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.B.2

The line plot shows the water in some beakers.
What is the total amount of water?



Total: _____

If all of the water were shared equally among the beakers, how much would each beaker hold?



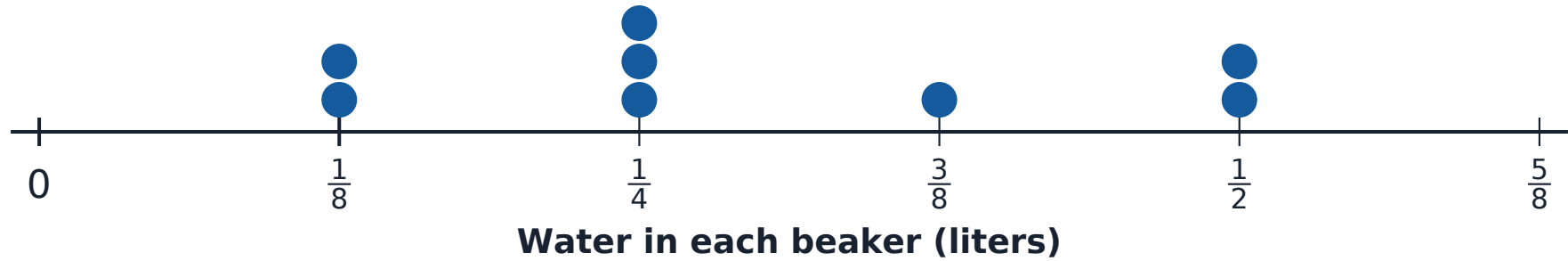
Liters: _____

How much more water is in the fullest beaker than in the least full beaker?



- A.** $\frac{1}{2}$ liter **B.** $\frac{1}{8}$ liter **C.** $\frac{5}{8}$ liter **D.** $\frac{3}{8}$ liter

What is the total amount of water in the beakers that hold $\frac{1}{4}$ liter each?



Total: _____

True or false?

The beakers that hold $\frac{1}{2}$ liter each contain 1 liter in all.



A. True

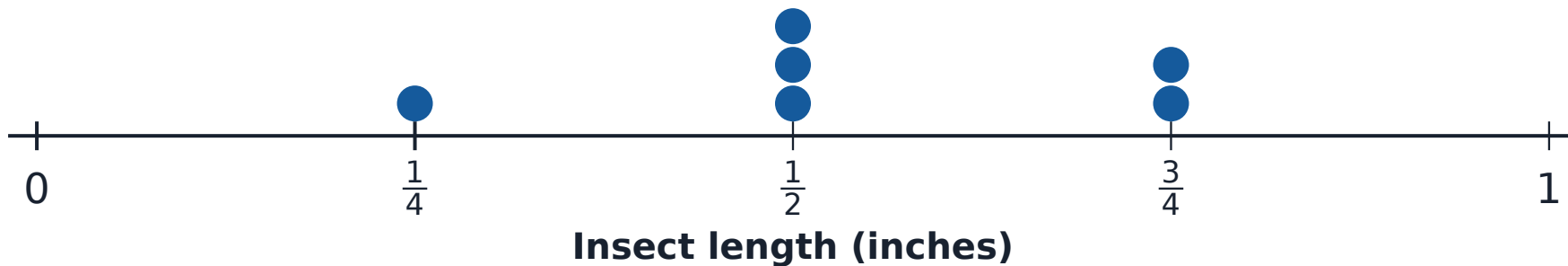
B. False

SET 44 • 5.MD.B.2

BACKWARD 1

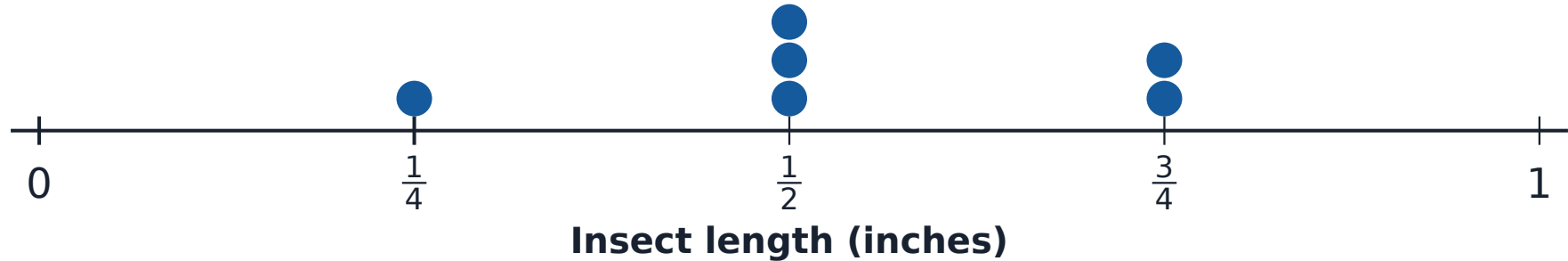
Solving problems from a line plot with like denominators • Grade 4 •
Measurement & Data • 4.MD.B.4

How many insects are $\frac{1}{2}$ inch long?



Insects: _____

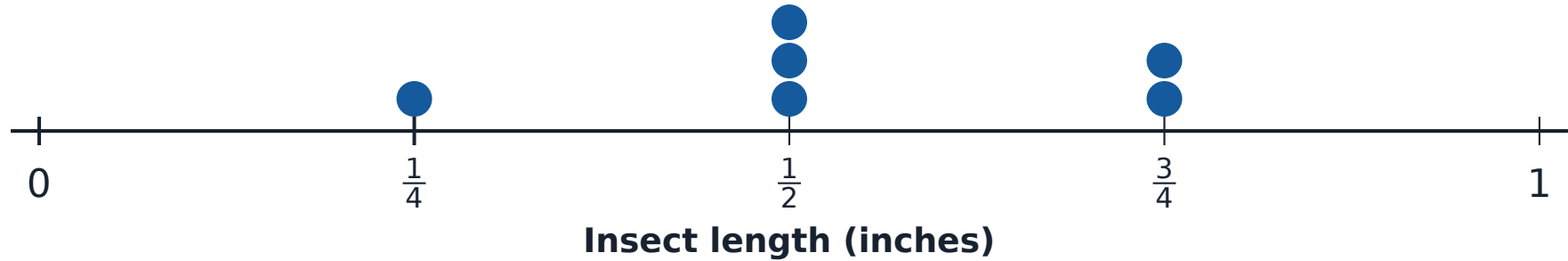
What is the difference between the longest and the shortest insect?



Difference: _____

True or false?

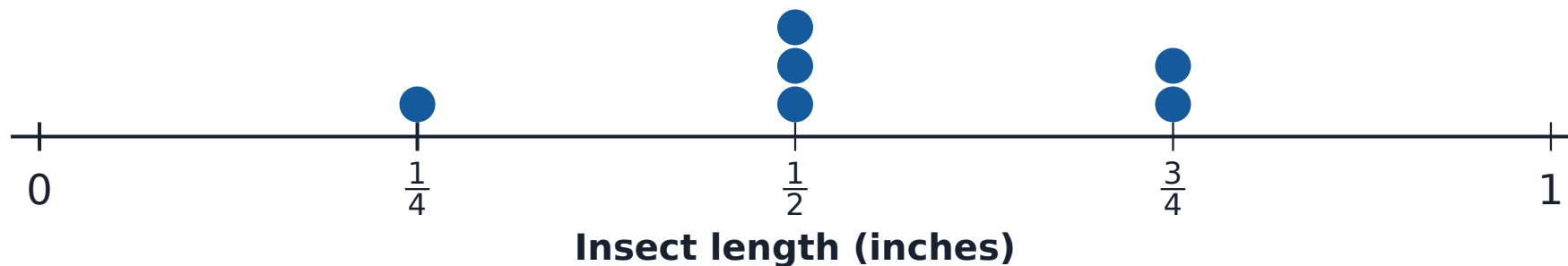
The total length of the $\frac{3}{4}$ -inch insects is $1\frac{1}{2}$ inches.



A. True

B. False

How many insects were measured?



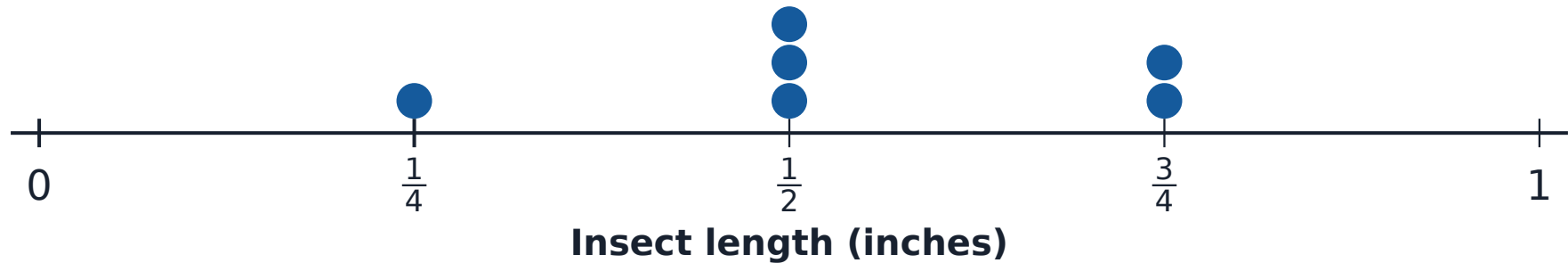
A. 4

B. 3

C. 5

D. 6

What is the total length of the $\frac{1}{2}$ -inch insects?



Total: _____

SET 44 • 5.MD.B.2

BACKWARD 4

Rename halves and fourths as eighths • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Write $\frac{1}{4}$ as eighths.

Fraction: _____

True or false?

$$\frac{1}{2} = \frac{4}{8}$$

A. True

B. False

Which fraction is equal to $\frac{3}{4}$?

A. $\frac{6}{8}$

B. $\frac{3}{8}$

C. $\frac{7}{8}$

D. $\frac{4}{8}$

Find the missing number.

$$\frac{1}{2} = \frac{?}{8}$$

Missing number: _____

True or false?

$$\frac{3}{8} = \frac{3}{4}$$

A. True

B. False

SET 44 • 5.MD.B.2

BACKWARD 5

Add fractions with like denominators • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.a

Add.

$$\frac{1}{8} + \frac{2}{8}$$

Sum: _____

True or false?

$$\frac{3}{8} + \frac{3}{8} = \frac{6}{16}$$

A. True

B. False

$$\frac{2}{8} + \frac{4}{8} + \frac{1}{8} = ?$$

A. $\frac{6}{8}$

B. $\frac{7}{8}$

C. $\frac{7}{24}$

D. 1

Add.

$$\frac{5}{8} + \frac{3}{8}$$

Sum: _____

True or false?

$$\frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{3}{8}$$

A. True

B. False

SET 44 • 5.MD.B.2

FORWARD 1

Find the mean of a data set • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean.

3, 5, 6, 8, 8

Mean: _____

Find the mean.

12, 15, 21

Mean: _____

Four friends have 2, 4, 5, and 9 marbles. They share the marbles equally.

How many marbles does each friend get?

A. 4.5

B. 5

C. 20

D. 4

The dot plot shows the goals scored in 6 games.
What is the mean number of goals?



Mean: _____

True or false?

The mean of 4, 4, and 10 is 4.

A. True

B. False

SET 44 • 5.MD.B.2

FORWARD 2

Compare two populations using means • Grade 7 • Statistics & Probability •
7.SP.B.4

Random samples: Sample A has a mean of 15 minutes.
Sample B has a mean of 22 minutes.

How much greater is the mean of Sample B?

Difference: _____

Two random samples of plant heights have means of 18 cm and 24 cm. Both samples have similar variability.

Which conclusion is best?

- A.** The populations are the same.
- B.** Plants in the second population tend to be taller.
- C.** Plants in the first population tend to be taller.
- D.** No conclusion is possible.

True or false?

Comparing the means of random samples is one way to compare two populations.

- A.** True
- B.** False

In random samples, Class X sleeps a mean of 7.5 hours and Class Y sleeps a mean of 8.25 hours.

Which class probably sleeps more?

Class: _____

A random sample of 30 students from School A spends a mean of 40 minutes on homework. A random sample of 30 students from School B spends a mean of 50 minutes. The samples have similar variability.

What can you infer about all the students at the two schools?

Inference: _____

Understand a unit cube and a cubic unit of volume

MAIN	Grade 5 main question	5.MD.C.3.a
BACKWARD 1	Unit squares and square units	3.MD.C.5.a
BACKWARD 2	Faces, edges, and vertices of a cube	2.G.A.1
FORWARD 1	Volume of cubes with fractional edge lengths	6.G.A.2
FORWARD 2	Volume of right prisms	7.G.B.6

SET 45 • 5.MD.C.3.a

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.3.a

True or false?

A cube with edges that are 1 unit long is called a unit cube.

- A.** True
- B.** False

What is the volume of a unit cube?

- A.** 1 square unit
- B.** 6 square units
- C.** 1 cubic unit
- D.** 3 cubic units

What is the volume of a cube with edges that are 1 cm long?

Volume: _____

Which object has a volume closest to 1 cubic inch?

- A.** A shoebox
- B.** A sugar cube
- C.** A bathtub
- D.** A school bus

True or false?

Volume measures the amount of space inside a solid figure.

A. True

B. False

SET 45 • 5.MD.C.3.a

BACKWARD 1

Unit squares and square units • Grade 3 • Measurement & Data • 3.MD.C.5.a

True or false?

A square with sides that are 1 unit long is called a unit square.

A. True

B. False

What is the area of a unit square?

- A.** 1 square unit
- B.** 1 cubic unit
- C.** 2 square units
- D.** 4 square units

How many unit squares cover a 3-by-3 square?

Unit squares: _____

True or false?

Area is measured in cubic units.

A. True

B. False

A square has sides that are 1 inch long.

What is its area?

Area: _____

SET 45 • 5.MD.C.3.a

BACKWARD 2

Faces, edges, and vertices of a cube • Grade 2 • Geometry • 2.G.A.1

How many faces does a cube have?

Faces: _____

True or false?

All the faces of a cube are squares.

A. True

B. False

How many edges does a cube have?

- A.** 6
- B.** 12
- C.** 4
- D.** 8

How many vertices (corners) does a cube have?

Vertices: _____

True or false?

All the edges of a cube are the same length.

A. True

B. False

SET 45 • 5.MD.C.3.a

FORWARD 1

Volume of cubes with fractional edge lengths • Grade 6 • Geometry • 6.G.A.2

What is the volume of a cube with edges that are $\frac{1}{2}$ inch long?

Volume: _____

How many cubes with $\frac{1}{2}$ -inch edges fit in a cube with 1-inch edges?

Cubes: _____

A cube has edges that are $\frac{1}{3}$ foot long.

What is its volume?

A. $\frac{1}{3}$ cubic foot

B. $\frac{1}{9}$ cubic foot

C. $\frac{1}{27}$ cubic foot

D. 1 cubic foot

A box is packed with 16 cubes that each have $\frac{1}{2}$ -cm edges.

What is the volume of the box?

Volume: _____

True or false?

Four cubes with $\frac{1}{2}$ -unit edges have a total volume of 1 cubic unit.

A. True

B. False

SET 45 • 5.MD.C.3.a

FORWARD 2

Volume of right prisms • Grade 7 • Geometry • 7.G.B.6

A prism has a base area of 24 square cm and a height of 5 cm.

What is its volume?

Volume: _____

A cube has edges that are 3 m long.

What is its volume?

Volume: _____

A triangular prism has a triangle base with an area of 6 square inches. The prism is 10 inches long.

What is its volume?

A. 60 cubic inches

B. 30 cubic inches

C. 16 cubic inches

D. 600 cubic inches

A box is 2.5 cm by 4 cm by 6 cm.

What is its volume?

Volume: _____

True or false?

Volume is measured in cubic units.

A. True

B. False

A solid packed with n unit cubes has a volume of n cubic units

MAIN

Grade 5 main question

5.MD.C.3.b

BACKWARD 1

Measuring area by counting unit squares

3.MD.C.6

BACKWARD 2

Arrays and repeated addition

2.OA.C.4

FORWARD 1

Pack prisms with fractional unit cubes

6.G.A.2

FORWARD 2

Solve real-world volume problems

7.G.B.6

SET 46 • 5.MD.C.3.b

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.3.b

A box is filled with 30 unit cubes with no gaps or overlaps.

What is its volume?

Volume: _____

True or false?

If unit cubes fill a box but leave gaps, the number of cubes is less than the volume of the box.

A. True

B. False

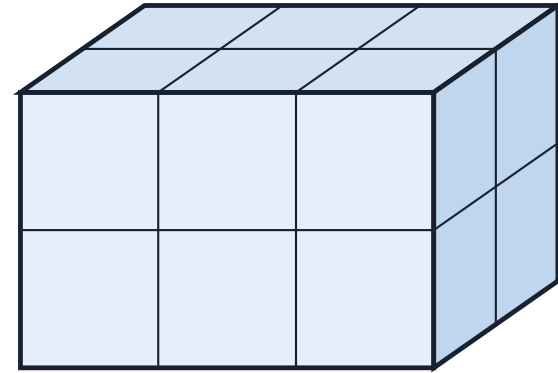
A solid is built from 18 unit cubes with no gaps or overlaps.

What is its volume?

- A.** 18 square units
- B.** 18 cubic units
- C.** 54 cubic units
- D.** 6 cubic units

Each cube is 1 cubic unit.

What is the volume of the solid?



Volume: _____

A tower is made of 9 cubes stacked with no gaps. Each cube is 1 cubic inch.

What is the volume of the tower?

Volume: _____

SET 46 • 5.MD.C.3.b

BACKWARD 1

Measuring area by counting unit squares • Grade 3 • Measurement & Data •
3.MD.C.6

A shape is covered by 12 unit squares with no gaps or overlaps.

What is its area?

Area: _____

True or false?

Overlapping squares give a wrong count for area.

A. True

B. False

A rectangle is covered by 20 unit squares.

What is its area?

- A.** 10 square units
- B.** 40 square units
- C.** 20 cubic units
- D.** 20 square units

Each square is 1 square cm. A rectangle is covered by 15 of these squares.

What is its area?

Area: _____

True or false?

Area can be found by counting the unit squares that cover a shape.

- A.** True
- B.** False

SET 46 • 5.MD.C.3.b

BACKWARD 2

Arrays and repeated addition • Grade 2 • Operations & Algebraic Thinking •
2.OA.C.4

An array has 4 rows with 5 in each row.

How many are there in all?

Total: _____

True or false?

3 rows of 5 make 8.

A. True

B. False

Which equation matches an array with 2 rows of 5?

A. $2 + 5 = 7$

B. $5 + 5 = 10$

C. $5 - 2 = 3$

D. $2 + 2 = 4$

An array has 5 rows with 3 in each row.

How many are there in all?

Total: _____

Write an addition equation for 3 rows of 4.

Equation: _____

SET 46 • 5.MD.C.3.b

FORWARD 1

Pack prisms with fractional unit cubes • Grade 6 • Geometry • 6.G.A.2

A box is packed with cubes that have $\frac{1}{2}$ -inch edges. It is 4 cubes long, 2 cubes wide, and 2 cubes tall.

What is the volume of the box?

Volume: _____

How many cubes with $\frac{1}{4}$ -inch edges fill a cube with 1-inch edges?

Cubes: _____

A prism is packed with 27 cubes that each have $\frac{1}{3}$ -foot edges.

What is its volume?

- A.** 27 cubic feet
- B.** 9 cubic feet
- C.** 1 cubic foot
- D.** 3 cubic feet

A box is $1\frac{1}{2}$ in. by 1 in. by $\frac{1}{2}$ in.

How many cubes with $\frac{1}{2}$ -inch edges fill it?

Cubes: _____

True or false?

40 cubes with $\frac{1}{2}$ -unit edges have a total volume of 20 cubic units.

- A. True
- B. False

SET 46 • 5.MD.C.3.b

FORWARD 2

Solve real-world volume problems • Grade 7 • Geometry • 7.G.B.6

A fish tank is 30 cm by 20 cm by 25 cm.
What is its volume?

Volume: _____

A prism has a volume of 96 cubic inches and a base area of 12 square inches.

What is its height?

Height: _____

A triangular prism has a triangle base with a base of 4 m and a height of 3 m. The prism is 7 m long.

What is its volume?

A. 84 cubic meters

B. 42 cubic meters

C. 21 cubic meters

D. 14 cubic meters

A storage box is 1.5 m by 1 m by 0.8 m.

What is its volume?

Volume: _____

True or false?

Two prisms with the same base area and the same height have the same volume.

A. True

B. False

Measure volume by counting unit cubes

MAIN	Grade 5 main question	5.MD.C.4
BACKWARD 1	Counting the squares in one layer	3.MD.C.7.a
BACKWARD 2	Equal groups	3.OA.A.1
FORWARD 1	Volume with fractional edge lengths	6.G.A.2
FORWARD 2	Volume of right prisms	7.G.B.6

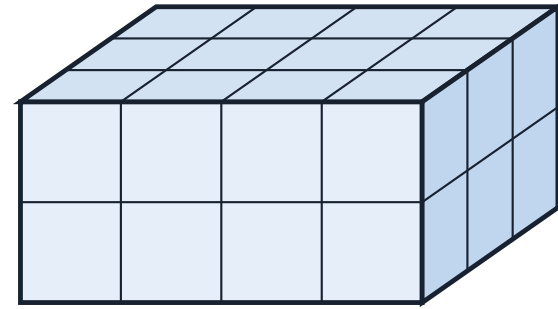
SET 47 • 5.MD.C.4

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.4

Each cube is 1 cubic cm.

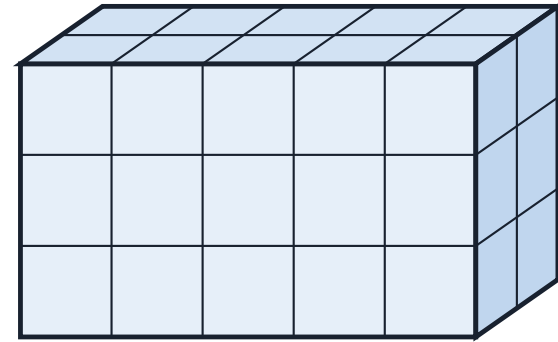
What is the volume of the prism?



Volume: _____

Each cube is 1 cubic inch.

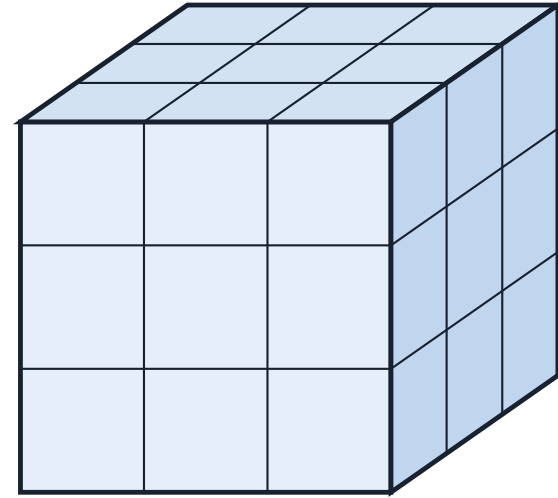
What is the volume of the prism?



Volume: _____

Each cube is 1 cubic foot.

What is the volume of the prism?



- A.** 18 cubic feet **B.** 54 cubic feet
- C.** 27 cubic feet **D.** 9 cubic feet

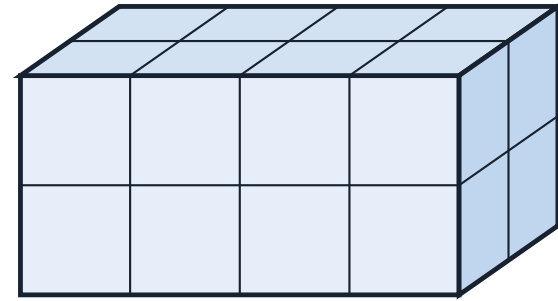
A box is filled with number cubes. The bottom layer has 6 rows of 4 cubes, and there are 3 layers.

What is the volume of the box, measured in number cubes?

Volume: _____

True or false?

Each cube is 1 cubic cm.
The volume of the prism is
16 cubic cm.



A. True

B. False

SET 47 • 5.MD.C.4

BACKWARD 1

Counting the squares in one layer • Grade 3 • Measurement & Data •
3.MD.C.7.a

The bottom layer of a box has 5 rows of 4 cubes.
How many cubes are in the layer?

Cubes: _____

True or false?

A layer with 3 rows of 6 cubes has 18 cubes.

A. True

B. False

How many unit squares cover a 4-by-6 rectangle?

- A.** 10
- B.** 46
- C.** 24
- D.** 20

A rectangle is tiled with 7 rows of 3 unit squares.
How many squares are there?

Squares: _____

True or false?

The number of unit squares that tile a rectangle is its length times its width.

- A. True
- B. False

SET 47 • 5.MD.C.4

BACKWARD 2

Equal groups • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

There are 3 layers with 12 cubes in each layer.
How many cubes are there in all?

Cubes: _____

True or false?

4 layers of 10 cubes is 14 cubes.

A. True

B. False

Which expression shows 5 layers of 8 cubes?

A. $8 - 5$

B. 5×8

C. $5 + 8$

D. 58

Multiply.

$$6 \times 9$$

Product: _____

There are 2 layers with 15 cubes in each layer.
How many cubes are there in all?

Cubes: _____

SET 47 • 5.MD.C.4

FORWARD 1

Volume with fractional edge lengths • Grade 6 • Geometry • 6.G.A.2

A prism is 3 cubes long, 2 cubes wide, and 2 cubes tall.
Each cube has $\frac{1}{2}$ -inch edges.

What is the volume of the prism?

Volume: _____

A box is $2\frac{1}{2}$ cm by 2 cm by 1 cm.

What is its volume?

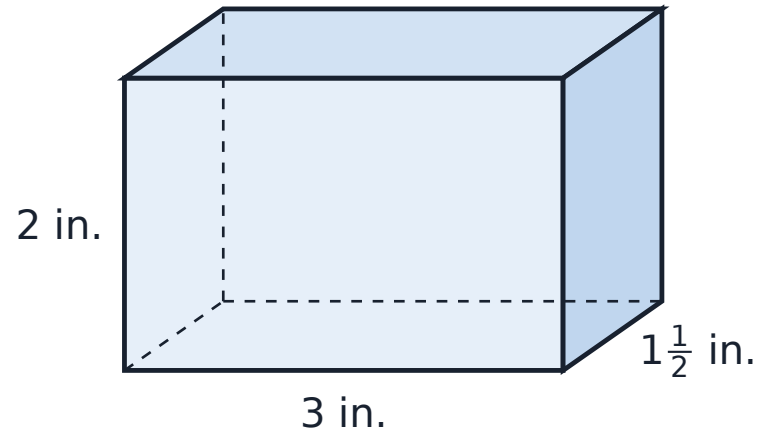
Volume: _____

How many cubes with $\frac{1}{2}$ -foot edges fill a box that is 2 ft by 1 ft by 1 ft?

- A.** 4
- B.** 8
- C.** 16
- D.** 2

Find the volume of the prism.

Volume: _____



True or false?

A cube with edges of $1\frac{1}{2}$ feet has a volume of $2\frac{1}{4}$ cubic feet.

- A.** True
- B.** False

SET 47 • 5.MD.C.4

FORWARD 2

Volume of right prisms • Grade 7 • Geometry • 7.G.B.6

A prism has a rectangular base that is 5 units by 4 units.
It is 6 units tall.

What is its volume?

Volume: _____

A box has a volume of 120 cubic cm. Its base is 6 cm by 5 cm.

What is its height?

Height: _____

A prism has a base area of 15 square feet and a height of 4 feet.

What is its volume?

- A.** 60 cubic feet
- B.** 30 cubic feet
- C.** 120 cubic feet
- D.** 19 cubic feet

A triangular prism has a base area of 9 square cm and a length of 12 cm.

What is its volume?

Volume: _____

True or false?

A box that is 10 in. by 2 in. by 3 in. has a volume of 15 cubic inches.

A. True

B. False

Relate packing with unit cubes to the formulas $V = l \times w \times h$ and $V = b \times h$

MAIN	Grade 5 main question	5.MD.C.5.a
BACKWARD 1	Area by tiling	3.MD.C.7.a
BACKWARD 2	Multiplying three numbers in any grouping	3.OA.B.5
FORWARD 1	Apply the volume formulas with fractional edge lengths	6.G.A.2
FORWARD 2	Volume of right prisms using $V = b \times h$	7.G.B.6

SET 48 • 5.MD.C.5.a

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.5.a

A prism is packed with unit cubes. It is 5 cubes long, 3 cubes wide, and 4 layers tall.

What is its volume?

Volume: _____

The bottom layer of a prism has 12 unit cubes. There are 5 layers.

Which expression gives the volume?

A. $12 \times 12 \times 5$

B. 12×5

C. 5×5

D. $12 + 5$

True or false?

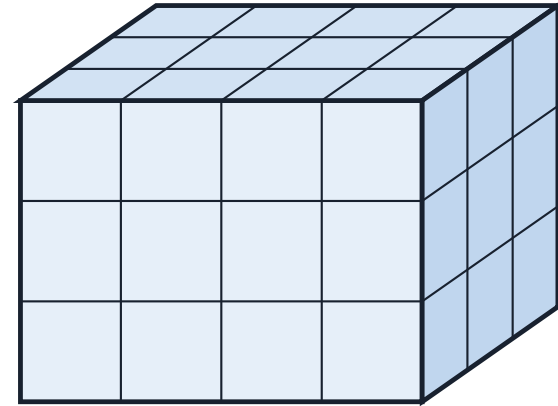
For a prism that is 4 by 3 by 2, $(4 \times 3) \times 2$ and $4 \times (3 \times 2)$ give the same volume.

A. True

B. False

Each cube is 1 cubic unit.

Find the volume by
multiplying the area of the
base by the height.



Volume: _____

A prism has a base area of 18 square units and a height of 4 units.

What is its volume?

Volume: _____

SET 48 • 5.MD.C.5.a

BACKWARD 1

Area by tiling • Grade 3 • Measurement & Data • 3.MD.C.7.a

A rectangle is tiled with unit squares in 3 rows of 4.
What is its area?

Area: _____

True or false?

The area of a 5-by-2 rectangle is 10 square units.

- A. True
- B. False

Which expression gives the area of a 6-by-3 rectangle?

A. $6 - 3$

B. $6 + 3$

C. 6×3

D. $2 \times (6 + 3)$

How many unit squares tile a 9-by-2 rectangle?

Squares: _____

True or false?

Tiling a rectangle and multiplying its side lengths give the same area.

A. True

B. False

SET 48 • 5.MD.C.5.a

BACKWARD 2

Multiplying three numbers in any grouping • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$(2 \times 3) \times 4 = 2 \times 3 + 4$$

A. True

B. False

Fill in the blank.

$$(5 \times 2) \times 3 = 5 \times (2 \times \underline{\quad})$$

Blank: _____

Which expression is equal to $4 \times 5 \times 2$?

A. 9×2

B. $20 + 2$

C. $4 + 5 + 2$

D. 4×10

Multiply.

$$3 \times 2 \times 5$$

Product: _____

True or false?

$$6 \times (2 \times 3) = 6 + 6$$

A. True

B. False

SET 48 • 5.MD.C.5.a

FORWARD 1

Apply the volume formulas with fractional edge lengths • Grade 6 • Geometry •
6.G.A.2

Use $V = l \times w \times h$ to find V when $l = 3\frac{1}{2}$ in., $w = 2$ in.,
and $h = 1\frac{1}{2}$ in.

Volume: _____

Use $V = b \times h$ to find the volume of a prism with $b = 6\frac{1}{4}$ square cm and $h = 4$ cm.

Volume: _____

A prism is packed with cubes that have edges $\frac{1}{2}$ unit long: 6 cubes long, 4 cubes wide, and 2 cubes tall.

Which shows its volume?

A. $6 + 4 + 2 = 12$ cubic units

B. $3 \times 2 \times 1 = 6$ cubic units

C. $6 \times 4 \times 2 = 48$ cubic units

D. $\frac{1}{2} \times 48 = 24$ cubic units

A cube has edges that are $\frac{3}{4}$ foot long.

What is its volume?

Volume: _____

True or false?

For a rectangular prism, $V = l \times w \times h$ and $V = b \times h$ give the same volume.

- A. True
- B. False

SET 48 • 5.MD.C.5.a

FORWARD 2

Volume of right prisms using $V = b \times h$ • Grade 7 • Geometry • 7.G.B.6

A triangular prism has a triangle base with a base of 6 cm and a height of 4 cm. The prism is 10 cm long.

What is its volume?

Volume: _____

A prism has a trapezoid base with an area of 14 square inches. The prism is 5 inches tall.

What is its volume?

Volume: _____

A prism has a volume of 84 cubic meters and a base area of 12 square meters.

What is its height?

A. 1,008 m

B. 7 m

C. 72 m

D. 96 m

A rectangular prism is 8 by 5 by 2.5.

What is its volume?

Volume: _____

True or false?

$V = b \times h$ works for triangular prisms as well as rectangular prisms.

A. True

B. False

Apply $V = l \times w \times h$ and $V = b \times h$ to solve real-world problems

MAIN	Grade 5 main question	5.MD.C.5.b
BACKWARD 1	Multiplying multi-digit numbers	4.NBT.B.5
BACKWARD 2	The area of the base	4.MD.A.3
FORWARD 1	Real-world volume with fractional edge lengths	6.G.A.2
FORWARD 2	Real-world volume problems with right prisms	7.G.B.6

SET 49 • 5.MD.C.5.b

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.5.b

A fish tank is 24 inches long, 12 inches wide, and 15 inches tall.

What is its volume?

Volume: _____

A sandbox has a base area of 30 square feet. It is filled with sand to a depth of 2 feet.

How much sand does it hold?

Volume: _____

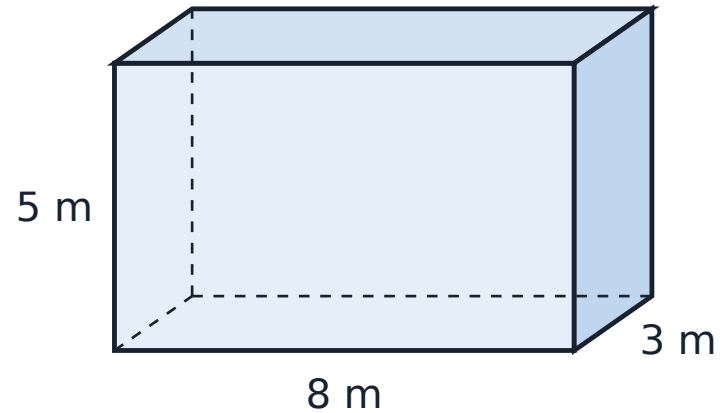
A box is 9 cm long, 4 cm wide, and 5 cm tall.

What is its volume?

- A.** 90 cubic cm
- B.** 36 cubic cm
- C.** 180 cubic cm
- D.** 18 cubic cm

Find the volume of the prism.

Volume: _____



A cube-shaped box has edges that are 7 inches long.
What is its volume?

Volume: _____

SET 49 • 5.MD.C.5.b

BACKWARD 1

Multiplying multi-digit numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

Multiply.

$$24 \times 12$$

Product: _____

Multiply.

$$288 \times 5$$

Product: _____

True or false?

$$36 \times 5 = 150$$

A. True

B. False

$$49 \times 7 = ?$$

- A.** 56
- B.** 343
- C.** 283
- D.** 353

Multiply.

$$15 \times 24$$

Product: _____

SET 49 • 5.MD.C.5.b

BACKWARD 2

The area of the base • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle is 24 in. by 12 in.

What is its area?

Area: _____

True or false?

A 9-by-4 rectangle has an area of 36 square units.

A. True

B. False

The base of a box is 8 m by 3 m.

What is the area of the base?

- A.** 11 square meters
- B.** 24 square meters
- C.** 22 square meters
- D.** 48 square meters

A square has sides that are 7 inches long.
What is its area?

Area: _____

A rectangle has an area of 30 square feet and a length of 6 feet.

What is its width?

Width: _____

SET 49 • 5.MD.C.5.b

FORWARD 1

Real-world volume with fractional edge lengths • Grade 6 • Geometry • 6.G.A.2

A box is $1\frac{1}{2}$ ft by 1 ft by $\frac{3}{4}$ ft.

What is its volume?

Volume: _____

A prism has a base area of $12\frac{1}{2}$ square inches and a height of 3 inches.

What is its volume?

Volume: _____

A cube has edges that are $2\frac{1}{2}$ cm long.

What is its volume?

A. $6\frac{1}{4}$ cubic cm

B. 15 cubic cm

C. $7\frac{1}{2}$ cubic cm

D. $15\frac{5}{8}$ cubic cm

A drawer is 2 ft by $1\frac{1}{2}$ ft by $\frac{1}{2}$ ft.

What is its volume?

Volume: _____

True or false?

A box that is 3 by $\frac{1}{2}$ by 4 has a volume of 12 cubic units.

A. True

B. False

SET 49 • 5.MD.C.5.b

FORWARD 2

Real-world volume problems with right prisms • Grade 7 • Geometry • 7.G.B.6

A tent is a triangular prism. Its triangle has a base of 2 m and a height of 1.5 m. The tent is 3 m long.

What is its volume?

Volume: _____

A pool is 10 m long, 5 m wide, and 2 m deep. It is filled to $\frac{3}{4}$ of its depth.

How much water is in the pool?

Volume: _____

A box has a volume of 360 cubic inches. Its base is 12 in. by 6 in.

What is its height?

A. 3 inches

B. 30 inches

C. 5 inches

D. 60 inches

A rectangular prism is 4.5 cm by 2 cm by 3 cm.
What is its volume?

Volume: _____

True or false?

A triangular prism with a base area of 8 square feet and a length of 5 feet holds 40 cubic feet.

A. True

B. False

Find a missing dimension when the volume is known

MAIN	Grade 5 main question	5.MD.C.5.b
BACKWARD 1	Dividing by a one-digit number	4.NBT.B.6
BACKWARD 2	Finding an unknown factor	3.OA.A.4
FORWARD 1	Solve one-step equations of the form $px = q$	6.EE.B.7
FORWARD 2	Solve two-step equations	7.EE.B.4.a

SET 50 • 5.MD.C.5.b

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.5.b

A box has a volume of 96 cubic inches. It is 6 inches long and 4 inches wide.

How tall is it?

Height: _____

A prism has a volume of 150 cubic cm and a height of 6 cm.

What is the area of its base?

Base area: _____

A tank holds 240 cubic feet. Its base is 8 feet by 5 feet.

How deep is the tank?

- A.** 6 feet
- B.** 30 feet
- C.** 40 feet
- D.** 227 feet

A rectangular prism has a volume of 84 cubic meters. It is 7 m long and 4 m tall.

How wide is it?

Width: _____

True or false?

A box with a volume of 60 cubic units and a base area of 12 square units is 5 units tall.

A. True

B. False

SET 50 • 5.MD.C.5.b

BACKWARD 1

Dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$96 \div 4$$

Quotient: _____

Divide.

$$150 \div 6$$

Quotient: _____

True or false?

$$240 \div 8 = 3$$

A. True

B. False

$$84 \div 7 = ?$$

A. 12

B. 77

C. 11

D. 13

Divide.

$$60 \div 5$$

Quotient: _____

SET 50 • 5.MD.C.5.b

BACKWARD 2

Finding an unknown factor • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.4

What number goes in the box?

$$6 \times \square = 48$$

Number: _____

True or false?

If $\square \times 5 = 35$, then $\square = 7$.

A. True

B. False

What number goes in the box?

$$4 \times \square = 36$$

- A.** 9
- B.** 8
- C.** 32
- D.** 40

What number goes in the box?

$$\square \times 7 = 63$$

Number: _____

True or false?

If $8 \times \square = 56$, then $\square = 6$.

A. True

B. False

SET 50 • 5.MD.C.5.b

FORWARD 1

Solve one-step equations of the form $px = q$ • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for h.

$$24h = 96$$

$$h = \underline{\hspace{10cm}}$$

Solve for b.

$$6b = 150$$

$$b = \underline{\hspace{10cm}}$$

Solve for w.

$$28w = 84$$

A. 56

B. 2,352

C. 112

D. 3

A prism has a volume of 52.5 cubic cm and a base area of 15 square cm.

Write and solve an equation to find its height h .

Equation and solution: _____

True or false?

The solution of $\frac{3}{4}x = 12$ is $x = 9$.

A. True

B. False

SET 50 • 5.MD.C.5.b

FORWARD 2

Solve two-step equations • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A box is 4 in. wide, 5 in. tall, and $(x + 2)$ in. long. Its volume is 120 cubic inches.

Find x .

$x =$ _____

Solve for x.

$$6(x + 3) = 54$$

x = _____

Solve for h.

$$12h + 8 = 68$$

A. 5

B. 6.3

C. 60

D. 4

Two boxes have a total volume of 100 cubic feet. One box holds 40 cubic feet. The other has a base area of 12 square feet.

Write and solve an equation to find the height h of the other box.

Equation and solution: _____

True or false?

The solution of $3(x - 2) = 21$ is $x = 9$.

A. True

B. False

Find the volume of a figure made of two rectangular prisms

MAIN	Grade 5 main question	5.MD.C.5.c
BACKWARD 1	Area of figures made of rectangles	3.MD.C.7.d
BACKWARD 3	The area formula for rectangles	4.MD.A.3
FORWARD 1	Volume of composite figures with fractional edge lengths	6.G.A.2
FORWARD 2	Volume of composite solids	7.G.B.6

SET 51 • 5.MD.C.5.c

MAIN

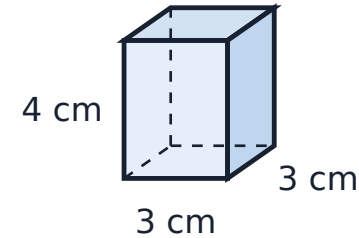
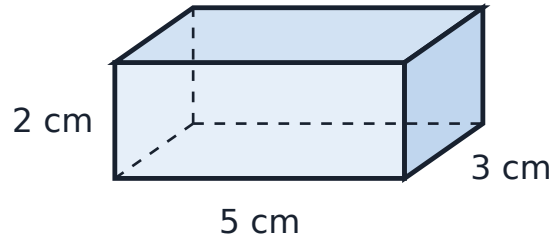
Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.5.c

A figure is made of two rectangular prisms that do not overlap. One is 6 ft by 4 ft by 3 ft. The other is 2 ft by 4 ft by 5 ft.

What is the total volume?

Volume: _____

A figure is made of the two prisms shown, which do not overlap.
What is the total volume?



Volume: _____

An L-shaped step is made of a 4-by-2-by-1 prism with a 2-by-2-by-1 prism on top.

What is its volume?

- A.** 12 cubic units
- B.** 8 cubic units
- C.** 10 cubic units
- D.** 16 cubic units

A building is a 10 m by 8 m by 6 m prism with a 4 m by 8 m by 3 m prism attached to one side.

What is the total volume?

Volume: _____

True or false?

The volume of a figure made of two prisms that do not overlap is the sum of their volumes.

- A. True
- B. False

SET 51 • 5.MD.C.5.c

BACKWARD 1

Area of figures made of rectangles • Grade 3 • Measurement & Data •
3.MD.C.7.d

A figure is split into two rectangles that do not overlap.
Their areas are 15 square units and 8 square units.

What is the area of the figure?

Area: _____

True or false?

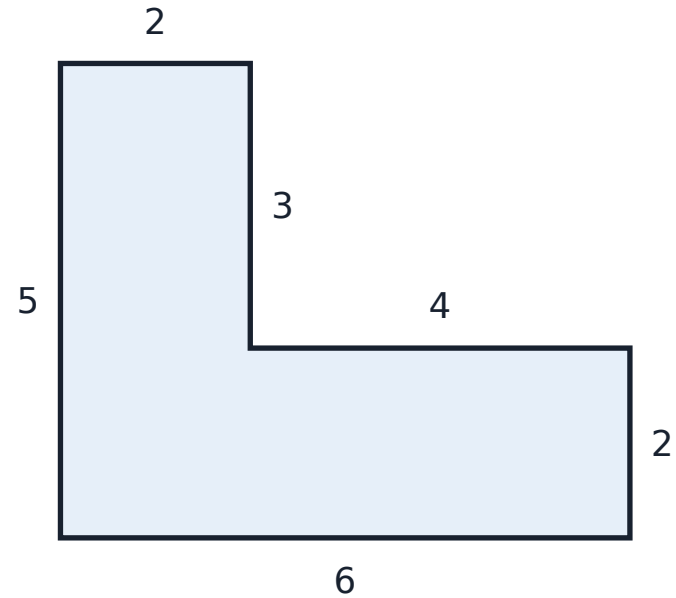
The area of a figure made of two non-overlapping parts is the sum of the areas of the parts.

- A. True
- B. False

The figure is split into a 6-by-2 rectangle along the bottom and a rectangle above it.

What is the area of the 6-by-2 rectangle?

- A.** 12 square units
- B.** 16 square units
- C.** 6 square units
- D.** 8 square units



A figure is made of two squares that do not overlap.
Their areas are 9 square units and 25 square units.

To find the area of the figure, what do you do with the two areas?

- A.** Divide them
- B.** Multiply them
- C.** Subtract them
- D.** Add them

A square with an area of 4 square units is cut out of a rectangle with an area of 48 square units.

What is the area of the part that is left?

Area: _____

SET 51 • 5.MD.C.5.c

BACKWARD 3

The area formula for rectangles • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle is 6 units by 4 units.

What is its area?

Area: _____

True or false?

A rectangle with an area of 32 square units and a length of 8 units has a width of 4 units.

A. True

B. False

A rectangle is 10 m by 8 m.

What is its area?

- A.** 36 square meters
- B.** 800 square meters
- C.** 80 square meters
- D.** 18 square meters

A rectangle has an area of 45 square feet and a width of 5 feet.

What is its length?

Length: _____

True or false?

A 4-by-8 rectangle has an area of 24 square units.

A. True

B. False

SET 51 • 5.MD.C.5.c

FORWARD 1

Volume of composite figures with fractional edge lengths • Grade 6 • Geometry
• 6.G.A.2

A figure is made of two prisms that do not overlap: $2\frac{1}{2}$ ft by 2 ft by 1 ft, and 1 ft by 2 ft by $\frac{1}{2}$ ft.

What is the total volume?

Volume: _____

Two prisms are joined. One has a volume of $4\frac{1}{2}$ cubic inches. The other is 3 in. by 1 in. by $\frac{1}{2}$ in.

What is the total volume?

Volume: _____

A step is made of a 3-by-1-by- $\frac{1}{2}$ prism and a 1-by-1-by- $\frac{1}{2}$ prism that do not overlap.

What is its volume?

A. 2 cubic units

B. $\frac{1}{2}$ cubic unit

C. 4 cubic units

D. $1\frac{1}{2}$ cubic units

A block that is 1 in. by 1 in. by $2\frac{1}{2}$ in. is cut out of a box that is 4 in. by 3 in. by $2\frac{1}{2}$ in.

What volume is left?

Volume: _____

True or false?

A figure made of two cubes with $\frac{1}{2}$ -foot edges has a volume of 1 cubic foot.

- A. True
- B. False

SET 51 • 5.MD.C.5.c

FORWARD 2

Volume of composite solids • Grade 7 • Geometry • 7.G.B.6

A figure is a 6-by-4-by-3 rectangular prism with a triangular prism on top. The triangle has a base of 4 and a height of 2, and the triangular prism is 6 long.

What is the total volume?

Volume: _____

A concrete step is made of two prisms: 1.5 m by 0.5 m by 0.2 m and 1.5 m by 0.25 m by 0.4 m.

What is the total volume?

Volume: _____

A 10-by-10-by-10 cube has a 2-by-2-by-10 hole cut all the way through it.

What volume remains?

- A.** 960 cubic units
- B.** 800 cubic units
- C.** 1,000 cubic units
- D.** 40 cubic units

A shed is a 3 m by 4 m by 2.5 m prism with a roof shaped like a triangular prism. The triangle has a base of 3 m and a height of 1 m, and the roof is 4 m long.

What is the total volume?

Volume: _____

True or false?

The volume of a composite solid can be found by adding the volumes of its parts.

A. True

B. False

Understand the coordinate plane: axes, origin, and coordinates

MAIN	Grade 5 main question	5.G.A.1
BACKWARD 1	Whole numbers on a number line	2.MD.B.6
BACKWARD 2	Perpendicular lines	4.G.A.1
FORWARD 1	Name the coordinates of points in all four quadrants	6.NS.C.6.c
FORWARD 2	Interpret a point on the graph of a proportional relationship	7.RP.A.2.d

SET 52 • 5.G.A.1

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.A.1

What is the point where the x-axis and the y-axis meet called?

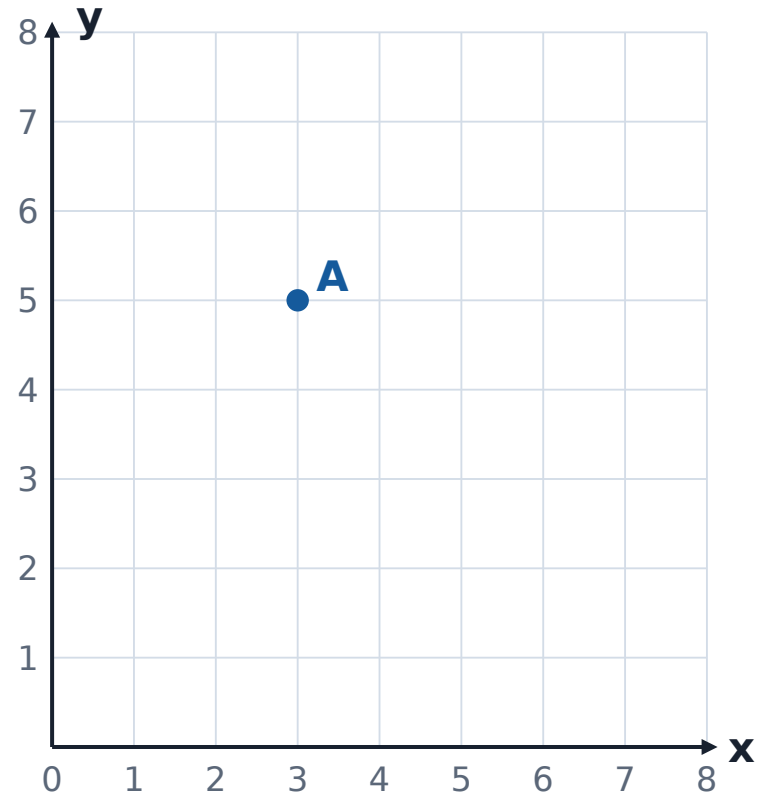
- A.** The y-axis
- B.** The origin
- C.** The x-coordinate
- D.** A quadrant

In the ordered pair $(7, 2)$, what does the 7 tell you?

Answer: _____

What are the coordinates of point A?

A = _____



True or false?

To reach $(0, 6)$, you start at the origin and move 6 units to the right along the x-axis.

- A. True
- B. False

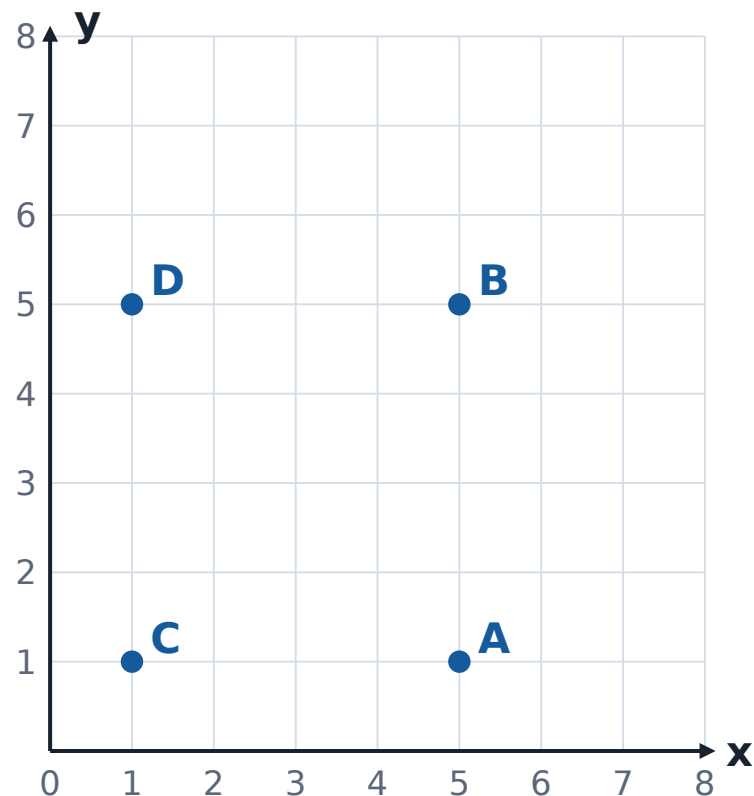
Which point is at (5, 1)?

A. Point A

B. Point B

C. Point C

D. Point D



SET 52 • 5.G.A.1

BACKWARD 1

Whole numbers on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

What number is at point P?



P = _____

True or false?

On a number line, numbers increase from left to right.

- A.** True
- B.** False

Which number is 4 units to the right of 0?

- A.** 4
- B.** 8
- C.** 40
- D.** 0

How many units is it from 0 to 9 on a number line?

Units: _____

True or false?

On a number line, 3 is farther from 0 than 7 is.

A. True

B. False

SET 52 • 5.G.A.1

BACKWARD 2

Perpendicular lines • Grade 4 • Geometry • 4.G.A.1

True or false?

Perpendicular lines meet to form right angles.

A. True

B. False

The x-axis and the y-axis are ____.

- A.** perpendicular
- B.** parallel
- C.** curved
- D.** the same line

How many degrees are in a right angle?

Degrees: _____

True or false?

Parallel lines never meet.

A. True

B. False

Which letter is made of two perpendicular line segments?

A. S

B. L

C. O

D. C

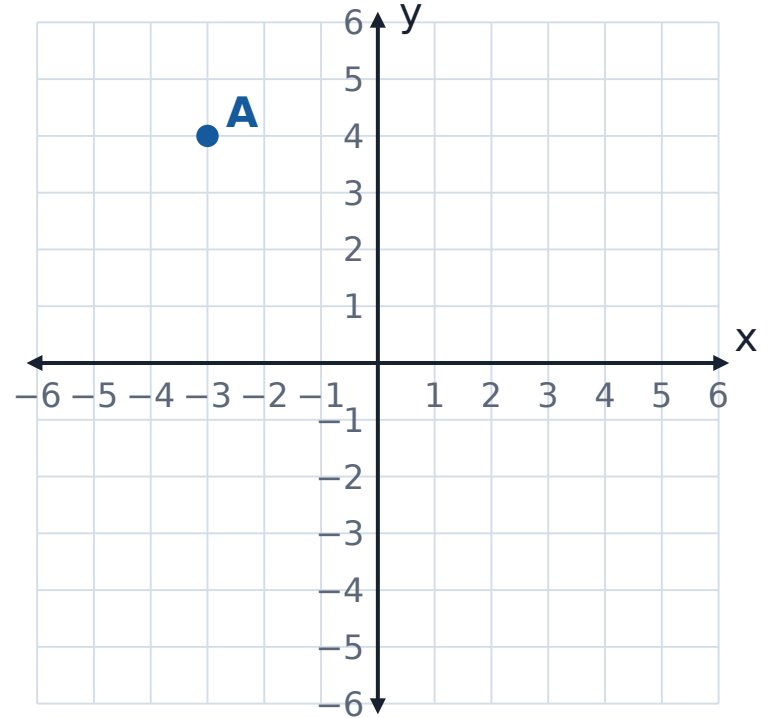
SET 52 • 5.G.A.1

FORWARD 1

Name the coordinates of points in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

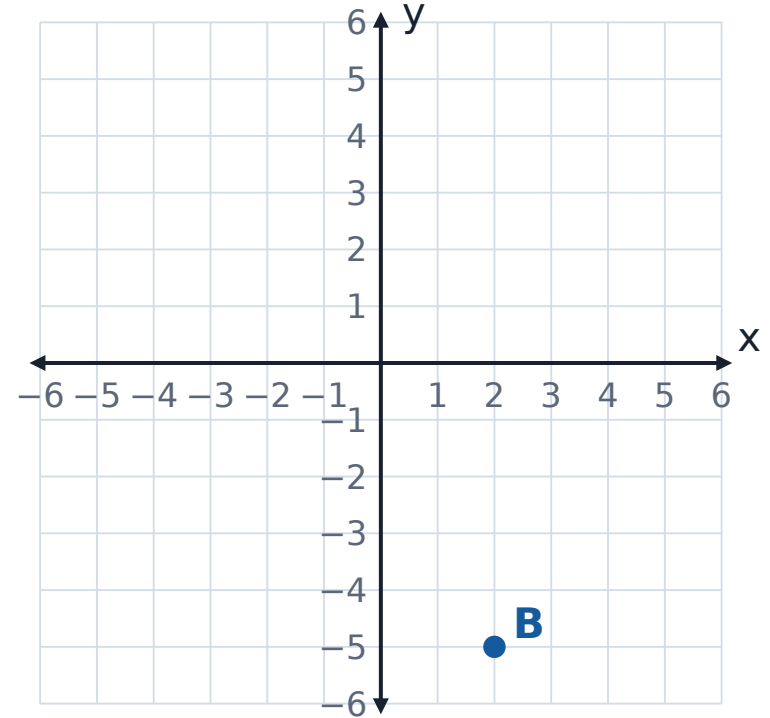
What are the coordinates of point A?

A = _____



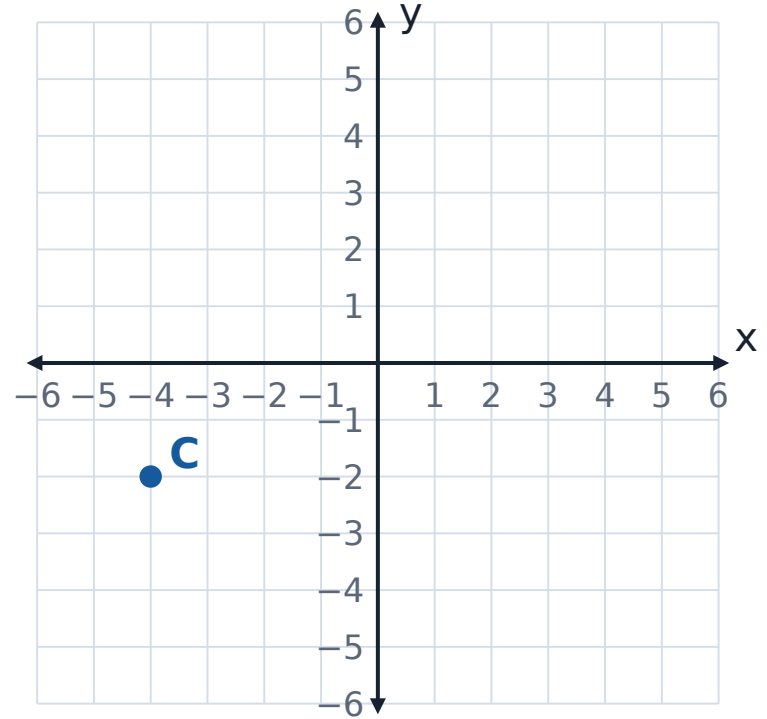
What are the coordinates of point B?

B = _____



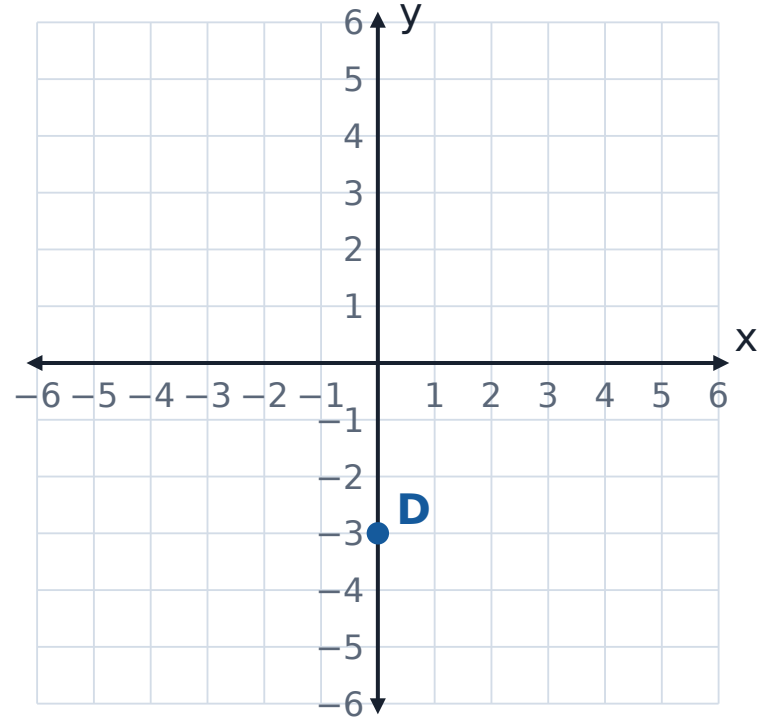
Which ordered pair names point C?

- A.** $(-4, -2)$
- B.** $(-2, -4)$
- C.** $(4, 2)$
- D.** $(-4, 2)$



What are the coordinates of point D?

D = _____

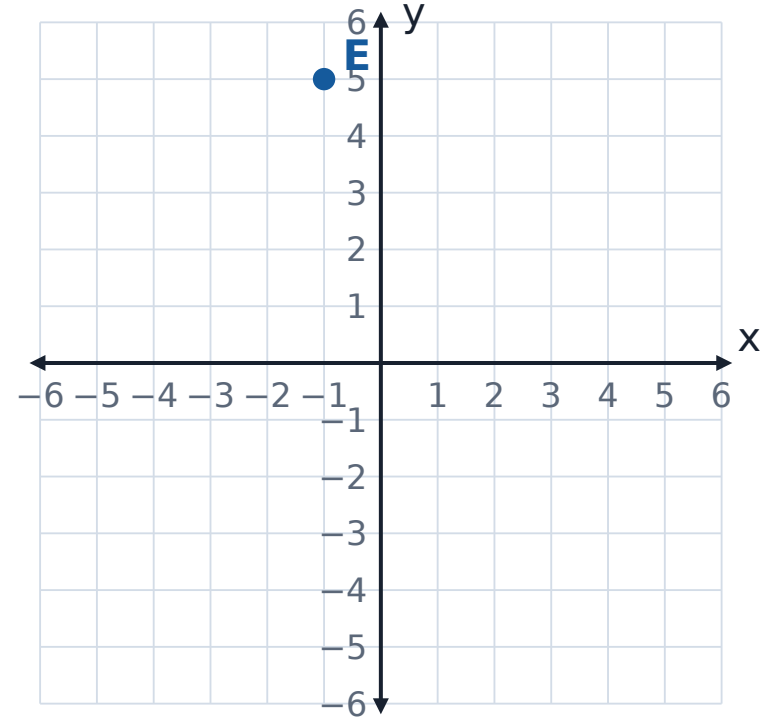


True or false?

Point E is located at $(5, -1)$.

A. True

B. False



SET 52 • 5.G.A.1

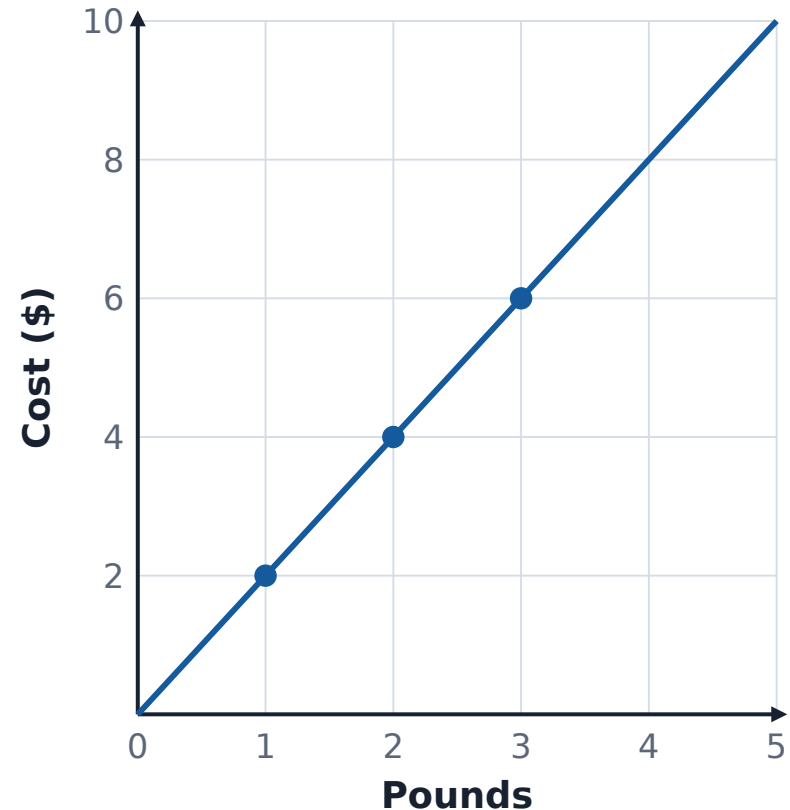
FORWARD 2

Interpret a point on the graph of a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d

The graph shows the cost of pears.

What does the point (3, 6) represent?

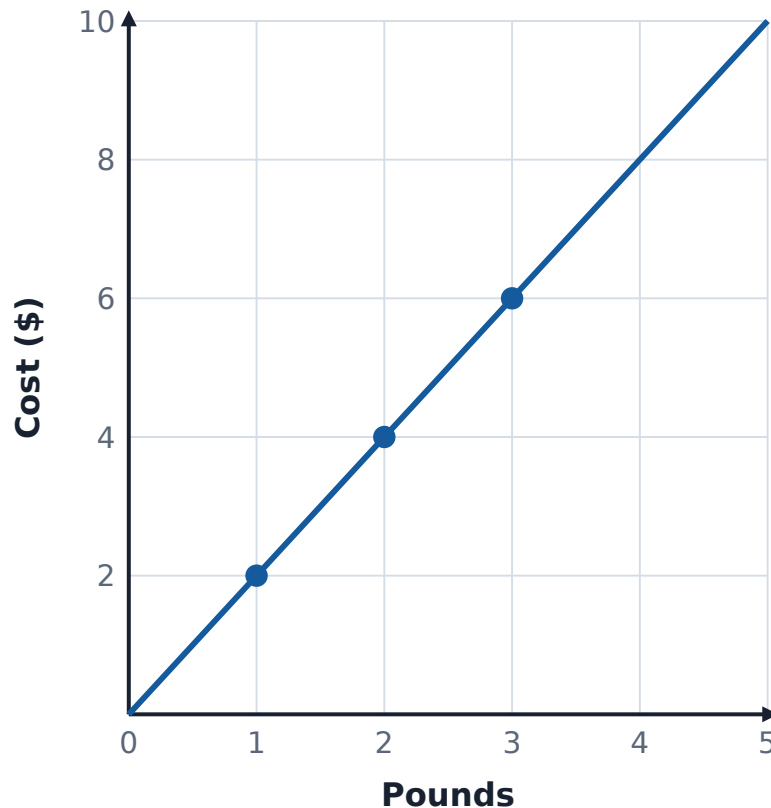
Meaning: _____



The graph shows the cost of pears.

What does the point (1, 2) represent?

- A.** The cost starts at \$2.
- B.** 2 pounds cost \$1.
- C.** \$1 buys 2 pounds.
- D.** 1 pound costs \$2.



True or false?

On the graph of a proportional relationship between pounds and cost, the point $(0, 0)$ means 0 pounds cost \$0.

- A. True
- B. False

The graph of a proportional relationship passes through (1, r) and (4, 20).

What is r ?

$r =$ _____

A graph shows a proportional relationship between the time x , in hours, that a bus drives and the distance y , in miles. It passes through $(2, 90)$.

What does this point mean?

- A.** The speed is 2 miles per hour.
- B.** In 90 hours, the distance is 2 miles.
- C.** The speed is 90 miles per hour.
- D.** In 2 hours, the distance is 90 miles.

Graph points in the first quadrant

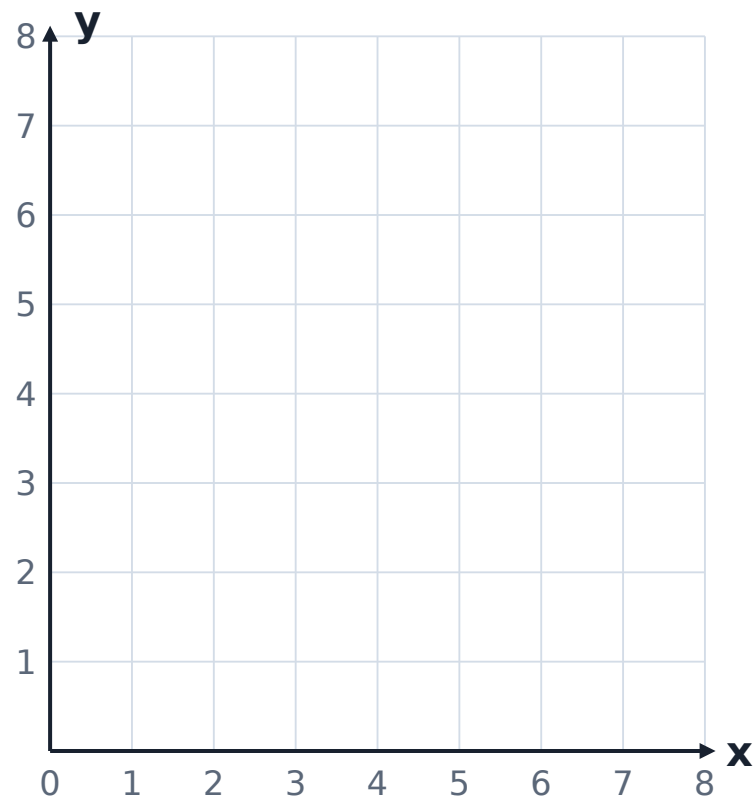
MAIN	Grade 5 main question	5.G.A.2
BACKWARD 2	Read a scaled bar graph	3.MD.B.3
BACKWARD 3	Fractions and whole numbers on a number line	3.NF.A.2
FORWARD 1	Reflect points across the axes	6.NS.C.6.b
FORWARD 2	Graph pairs to decide whether a relationship is proportional	7.RP.A.2.a

SET 53 • 5.G.A.2

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.A.2

Plot and label the points A(2, 7), B(6, 3), and C(0, 5).

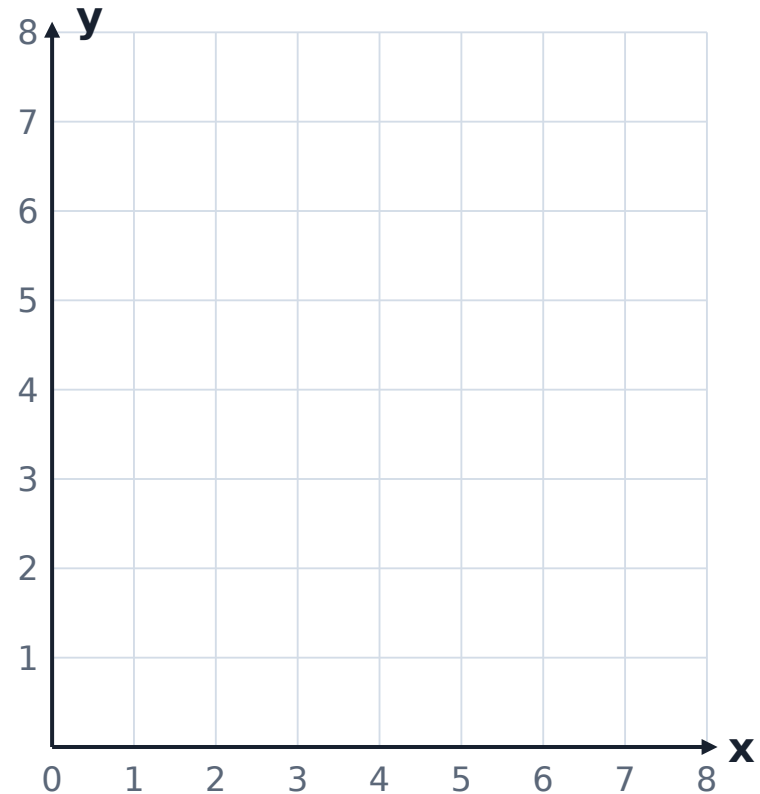


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q1

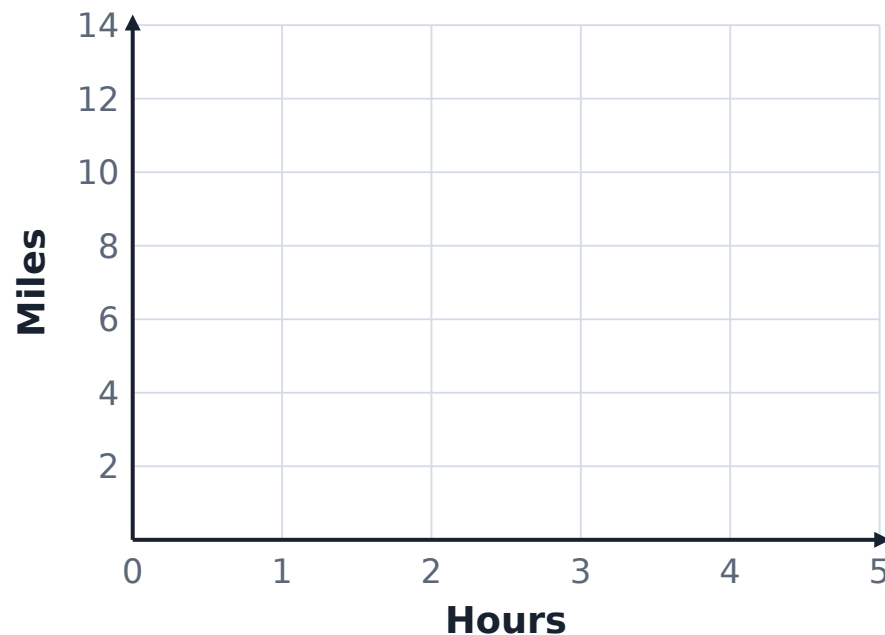
Grade 5 • Geometry • 5.G.A.2 • Set 53 • MAIN • Q1/5

Plot and label the points D(4, 4), E(7, 0), and F(1, 6).

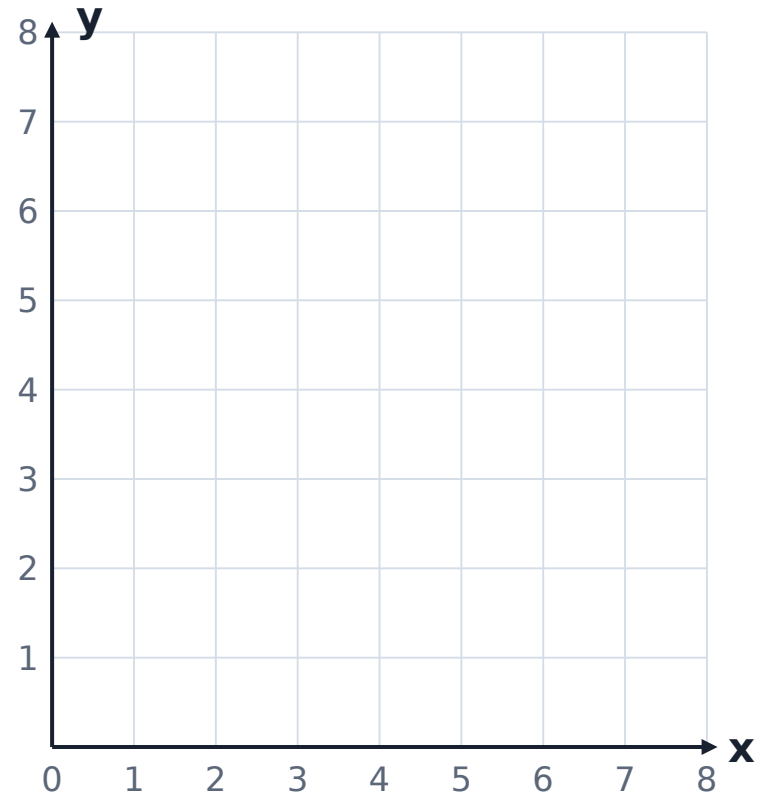


Plot the points from the table.

Hours	1	2	3	4
Miles	3	6	9	12



Plot and label the points $G(3, 2\frac{1}{2})$ and $H(5\frac{1}{2}, 4)$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

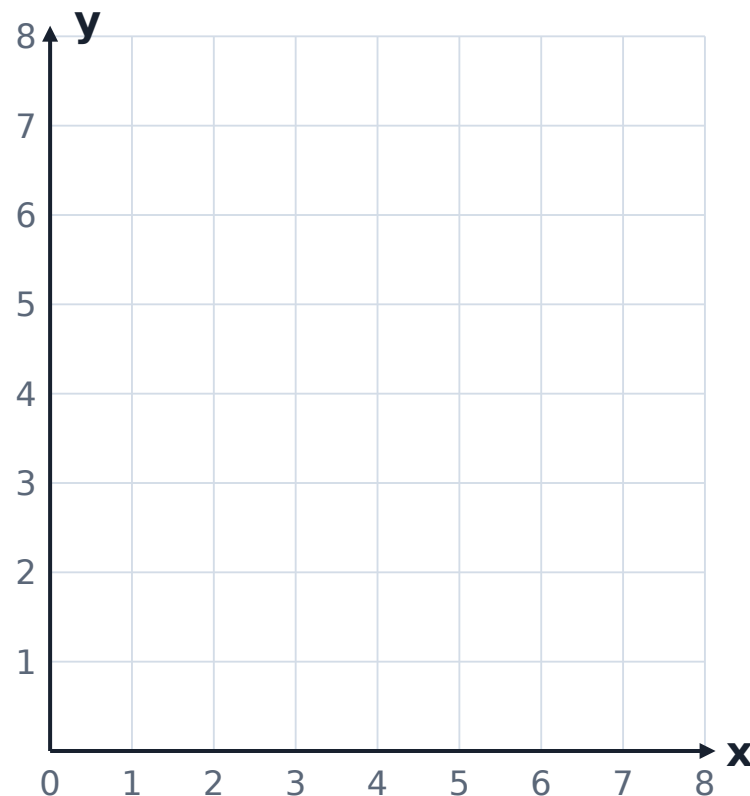
ID S53-M-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 53 • MAIN • Q4/5

Plot $P(2, 3)$ and $Q(2, 7)$.

How far apart are the two points?

Distance: _____

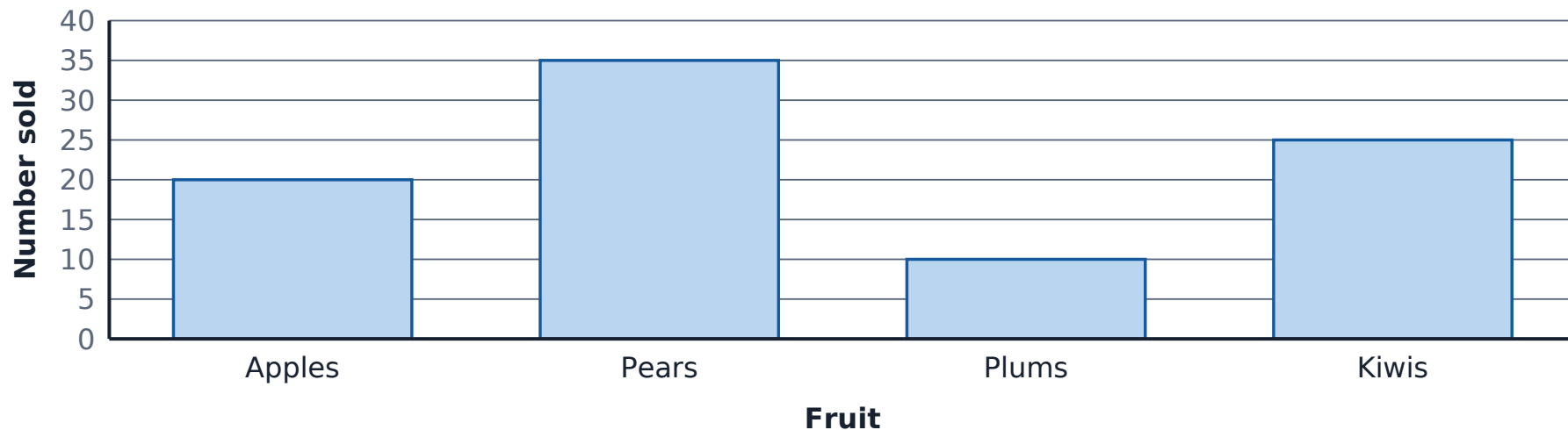


SET 53 • 5.G.A.2

BACKWARD 2

Read a scaled bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

What does each grid line on the scale stand for?



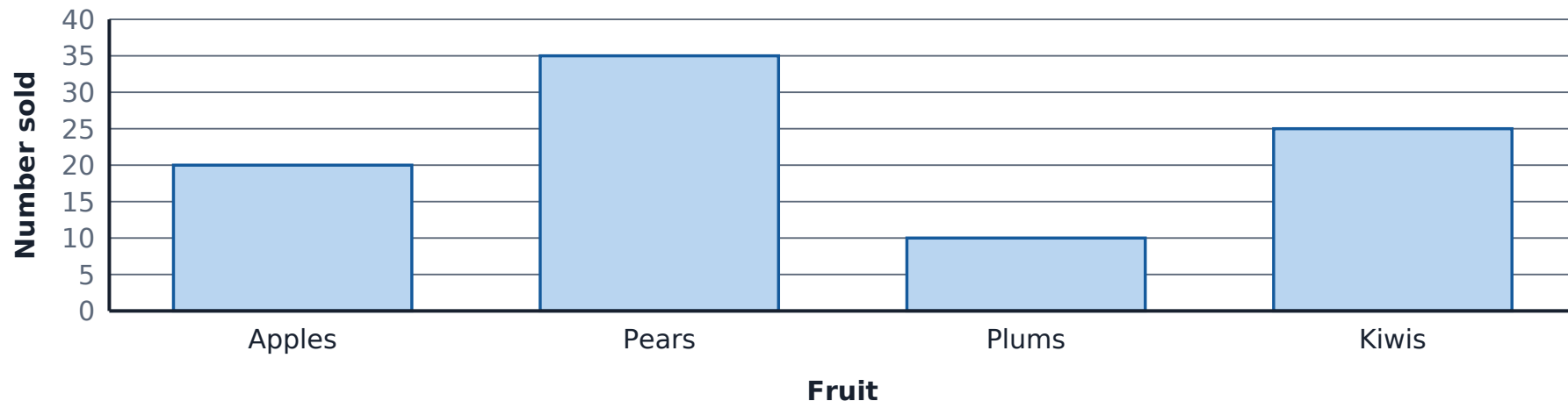
A. 10

B. 20

C. 5

D. 1

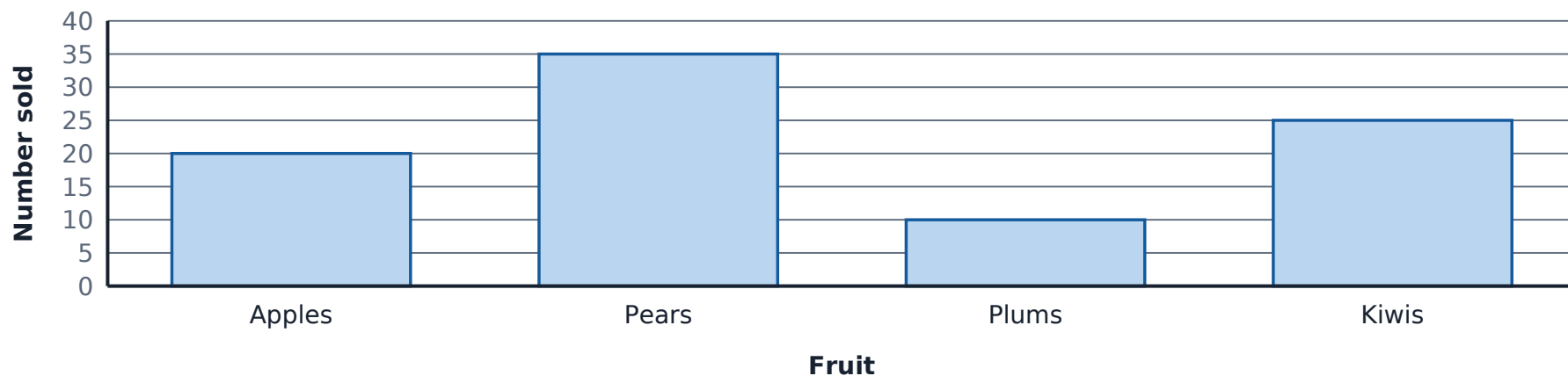
How many pears were sold?



Pears: _____

True or false?

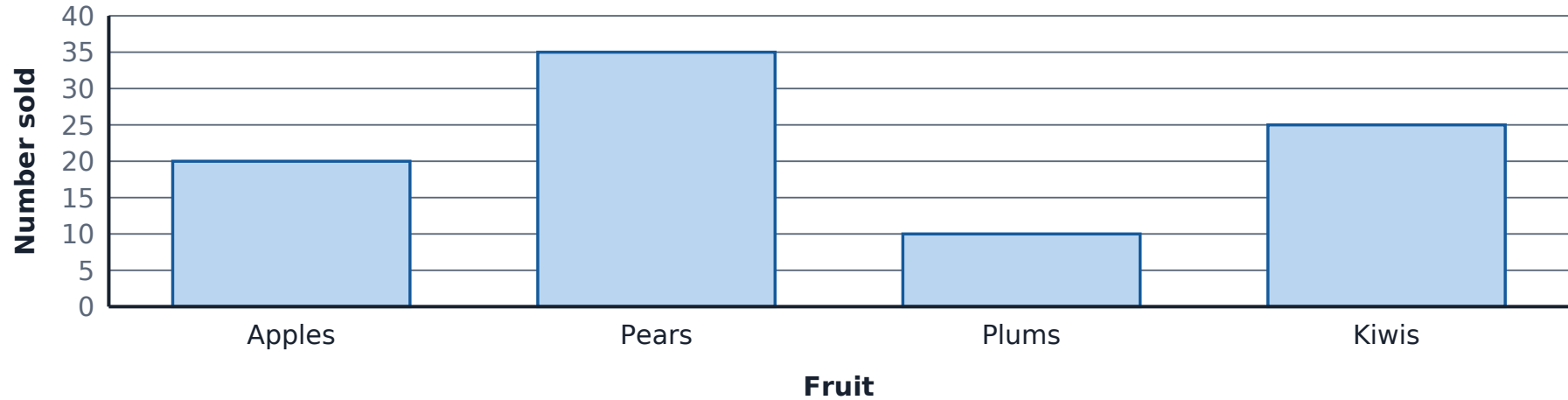
More kiwis than apples were sold.



A. True

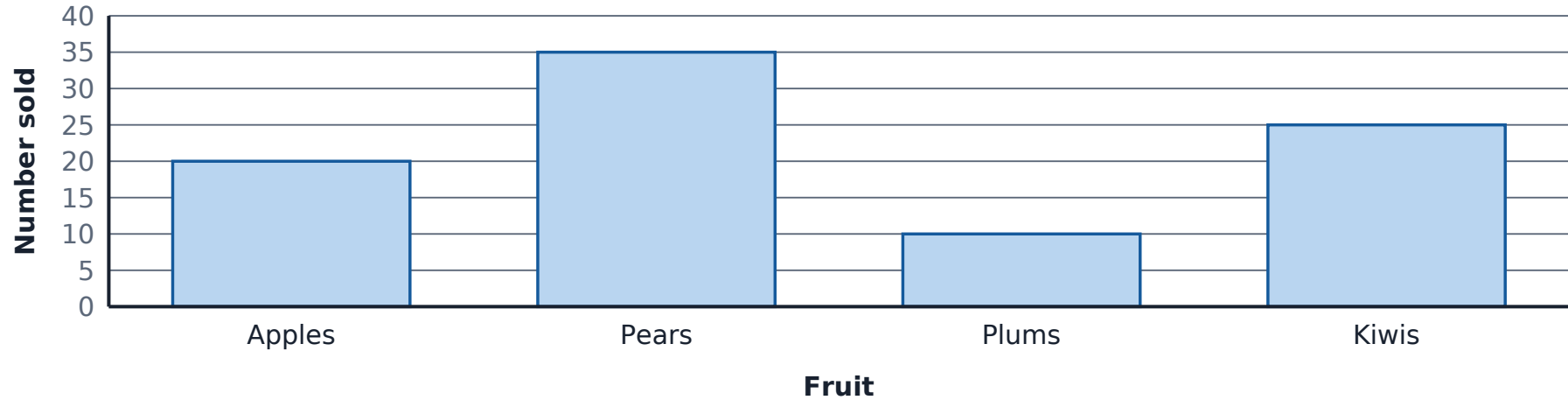
B. False

How many fewer plums than pears were sold?



Plums: _____

How many apples and kiwis were sold in all?



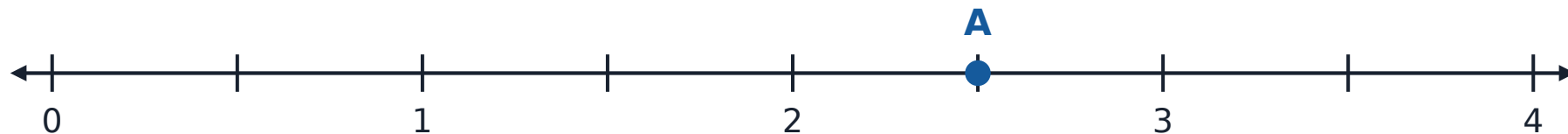
Fruit: _____

SET 53 • 5.G.A.2

BACKWARD 3

Fractions and whole numbers on a number line • Grade 3 • Number & Operations–Fractions • 3.NF.A.2

What number is at point A?



A = _____

True or false?

$2\frac{1}{2}$ is halfway between 2 and 3 on a number line.

A. True

B. False

Which point is at $1\frac{1}{2}$?



A. Point A

B. Point B

C. Point C

D. Point D

What number is at point B?



B = _____

True or false?

On a number line, $5\frac{1}{2}$ is between 6 and 7.

A. True

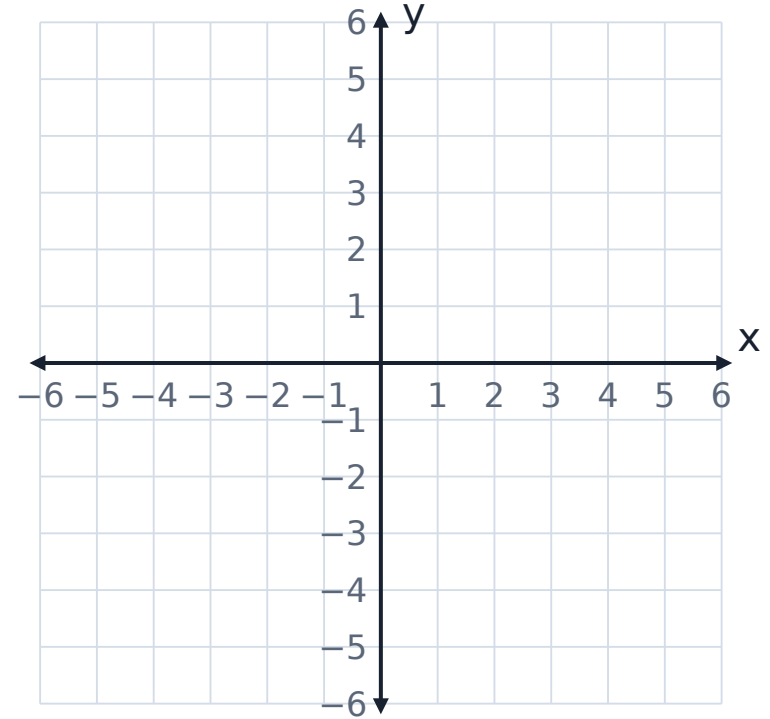
B. False

SET 53 • 5.G.A.2

FORWARD 1

Reflect points across the axes • Grade 6 • The Number System • 6.NS.C.6.b

Plot and label A(2, 4). Then plot its reflection across the x-axis and label it B.



The point $(5, 2)$ is reflected across the x -axis.
What are the coordinates of its image?

Image: _____

The point $(-3, 5)$ is reflected across the x -axis.
What are the coordinates of its image?

- A.** $(3, -5)$
- B.** $(3, 5)$
- C.** $(-3, -5)$
- D.** $(5, -3)$

The point $(-4, 1)$ is reflected across the y -axis.
What are the coordinates of its image?

Image: _____

True or false?

The points $(3, 5)$ and $(-3, 5)$ are reflections of each other across the x-axis.

A. True

B. False

SET 53 • 5.G.A.2

FORWARD 2

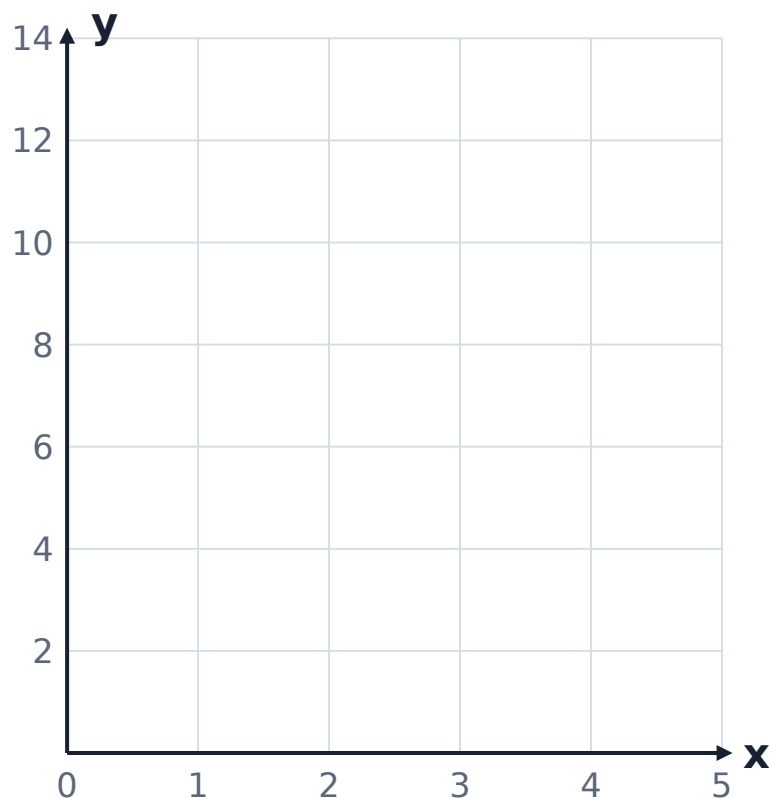
Graph pairs to decide whether a relationship is proportional • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

Graph the pairs $(1, 3)$, $(2, 6)$, and $(4, 12)$.

Is the relationship proportional? Explain using your graph.

Answer: _____

Reason: _____

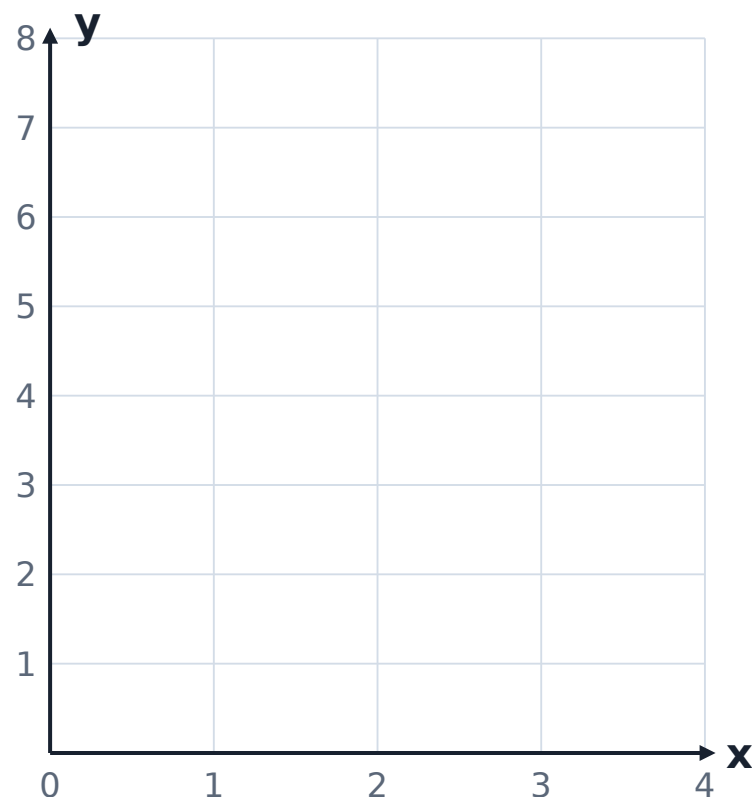


Graph the pairs $(0, 2)$, $(1, 4)$, and $(2, 6)$.

Is the relationship proportional? Explain using your graph.

Answer: _____

Reason: _____



Which graph shows a proportional relationship?

- A.** A curve through $(0, 0)$
- B.** Points that are not on a line
- C.** A straight line through $(0, 3)$
- D.** A straight line through $(0, 0)$

True or false?

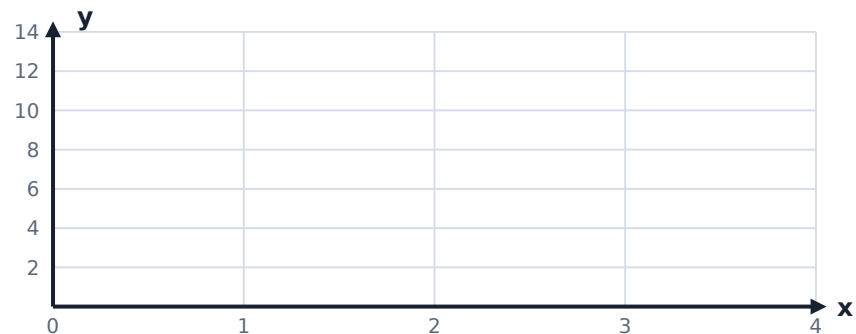
The points (2, 1), (4, 2), and (6, 3) show a proportional relationship.

- A. True
- B. False

Graph the pairs from the table.

Is the relationship proportional? Explain using your graph.

x	1	2	3
y	4	8	12



Answer: _____

Reason: _____

Solve real-world problems by graphing points and interpreting coordinates

MAIN	Grade 5 main question	5.G.A.2
BACKWARD 2	Distance between whole numbers on a number line	2.MD.B.6
BACKWARD 3	Solve problems with a scaled bar graph	3.MD.B.3
FORWARD 1	Find distances in all four quadrants	6.NS.C.8
FORWARD 2	Use a scale on a map grid	7.G.A.1

SET 54 • 5.G.A.2

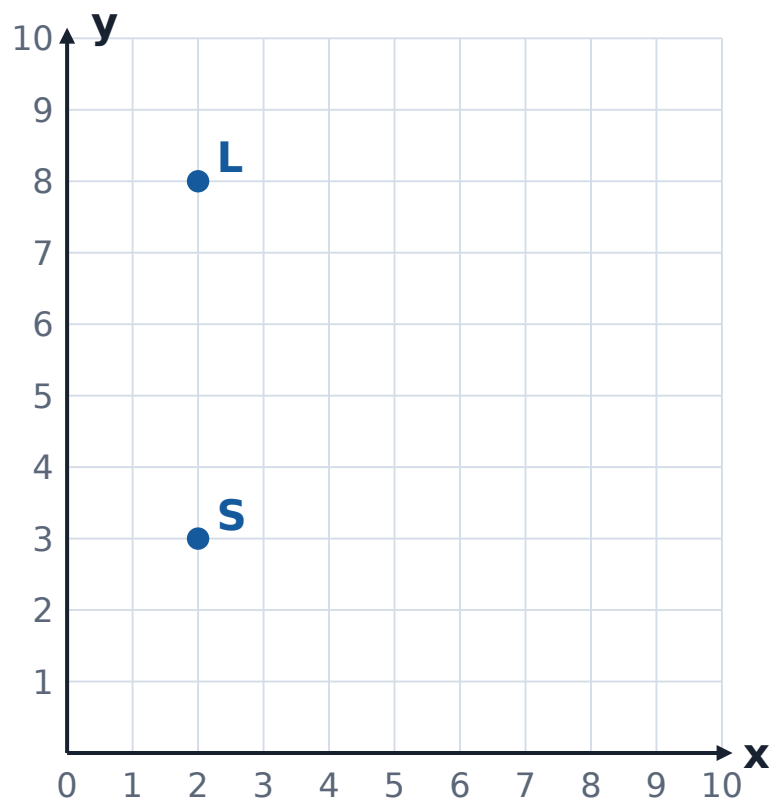
MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.A.2

The grid is a map. Each unit is 1 block.

How many blocks is it from the school S to the library L?

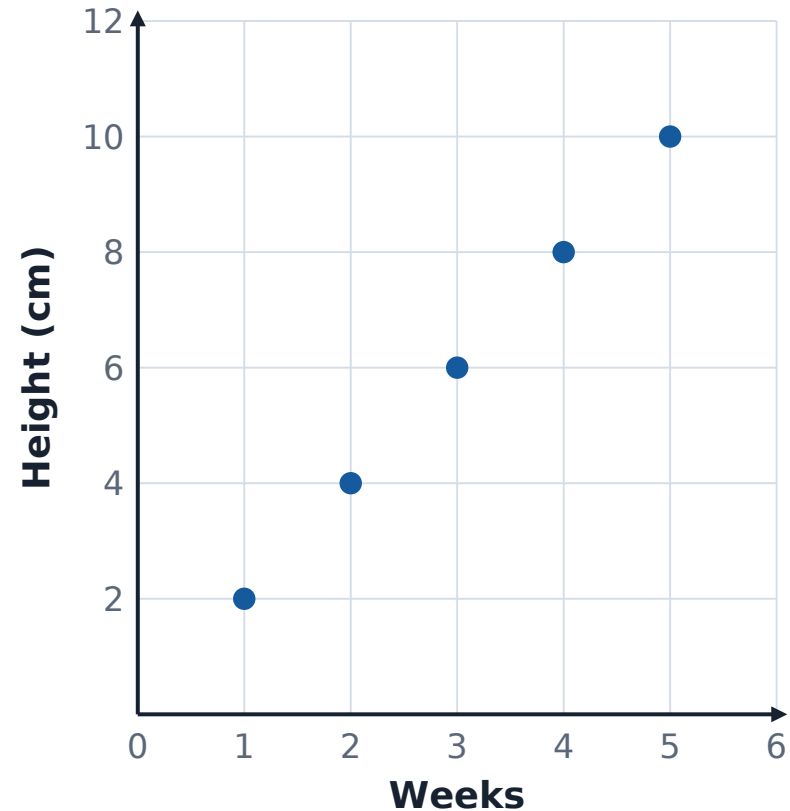
Blocks: _____



The graph shows the height of a plant.

How tall was the plant after 4 weeks?

Height: _____



The point (3, 12) is on a graph of hours worked (x) and dollars earned (y).

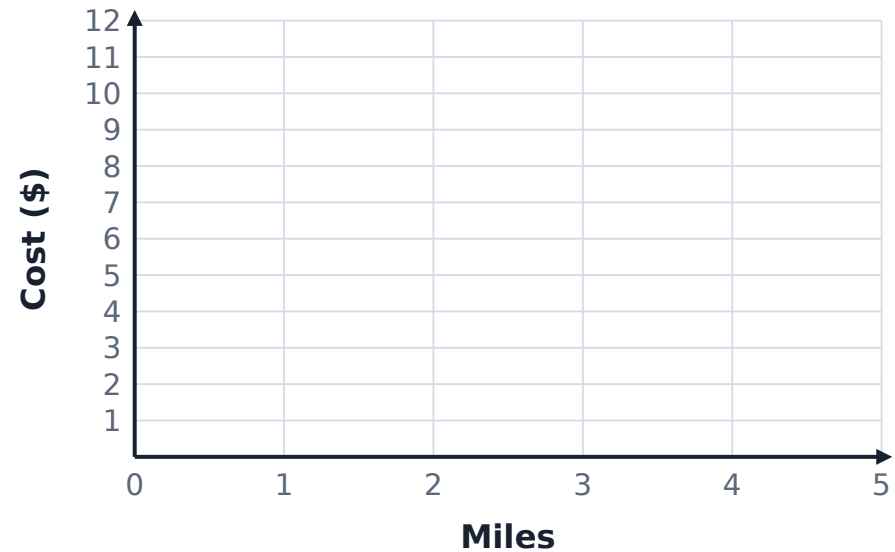
What does the point mean?

- A.** Working 3 hours earns \$12.
- B.** Working 12 hours earns \$3.
- C.** The pay starts at \$12.
- D.** Each hour earns \$3.

The table shows the cost of a taxi ride.

Graph the points.

Miles	1	2	3	4
Cost (\$)	5	7	9	11



On a map grid, the park is at (1, 6) and the pool is at (7, 6). Each unit is 100 meters.

How far apart are the park and the pool?

Distance: _____

SET 54 • 5.G.A.2

BACKWARD 2

Distance between whole numbers on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

What is the distance from 2 to 7 on a number line?

Distance: _____

True or false?

The distance from 3 to 10 is 13 units.

A. True

B. False

How many units apart are 1 and 8 on a number line?

A. 9

B. 8

C. 1

D. 7

What is the distance from 6 to 6 on a number line?

Distance: _____

How many units apart are points A and B?



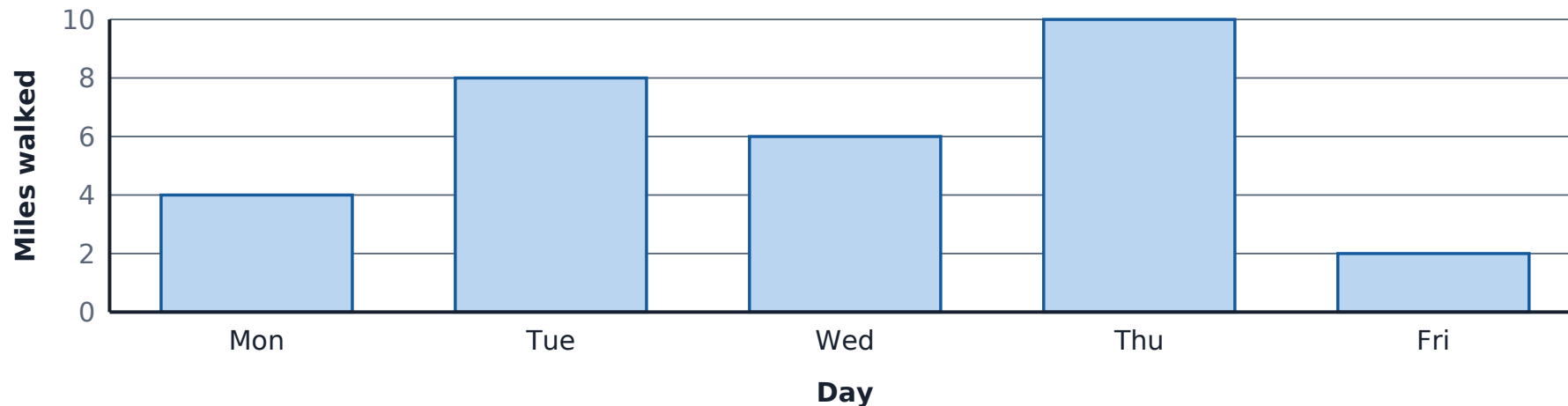
Units: _____

SET 54 • 5.G.A.2

BACKWARD 3

Solve problems with a scaled bar graph • Grade 3 • Measurement & Data •
3.MD.B.3

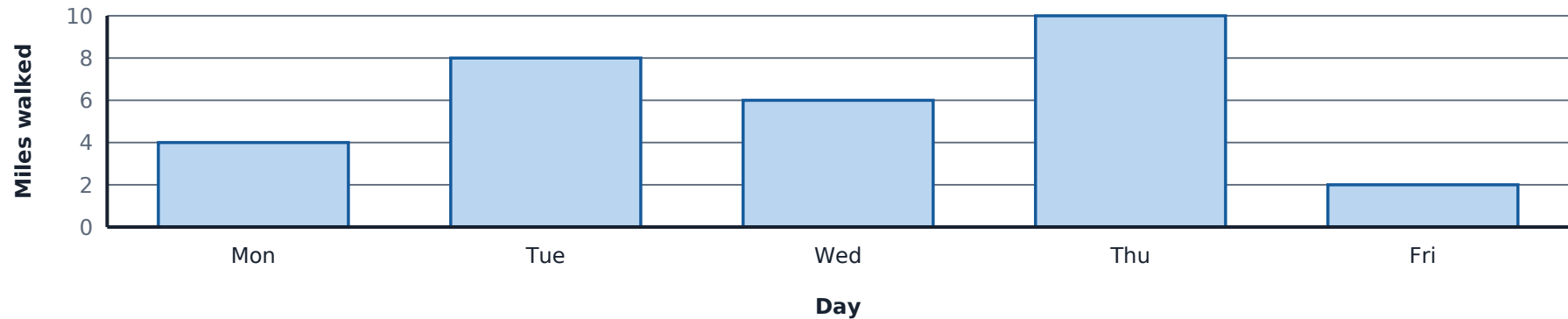
How many miles were walked on Thursday?



Miles: _____

True or false?

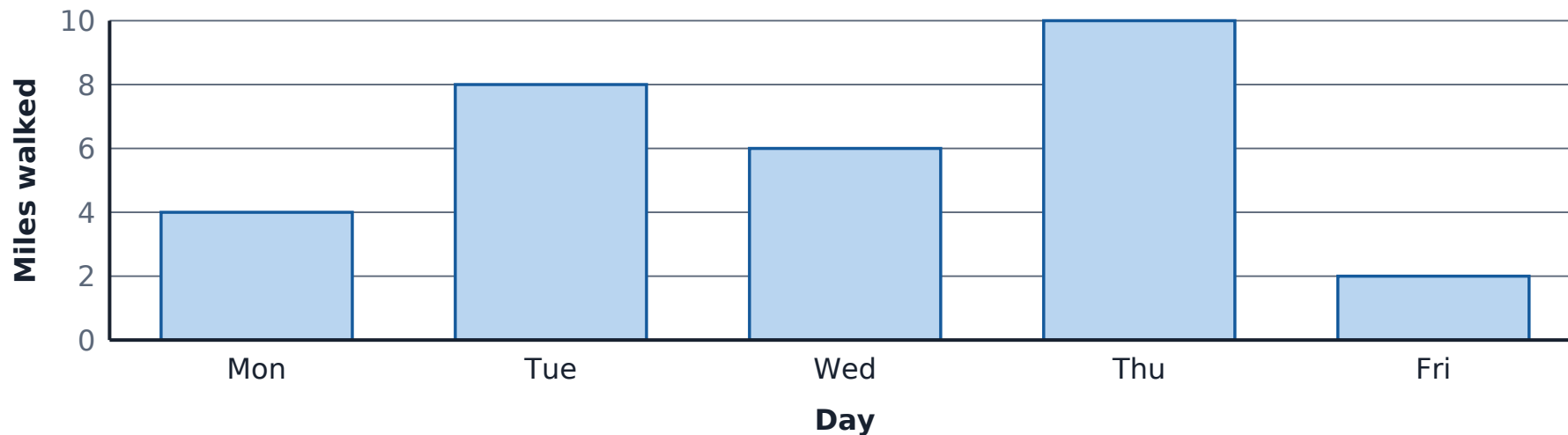
The same number of miles were walked on Monday and Wednesday.



A. True

B. False

On which day were 8 miles walked?



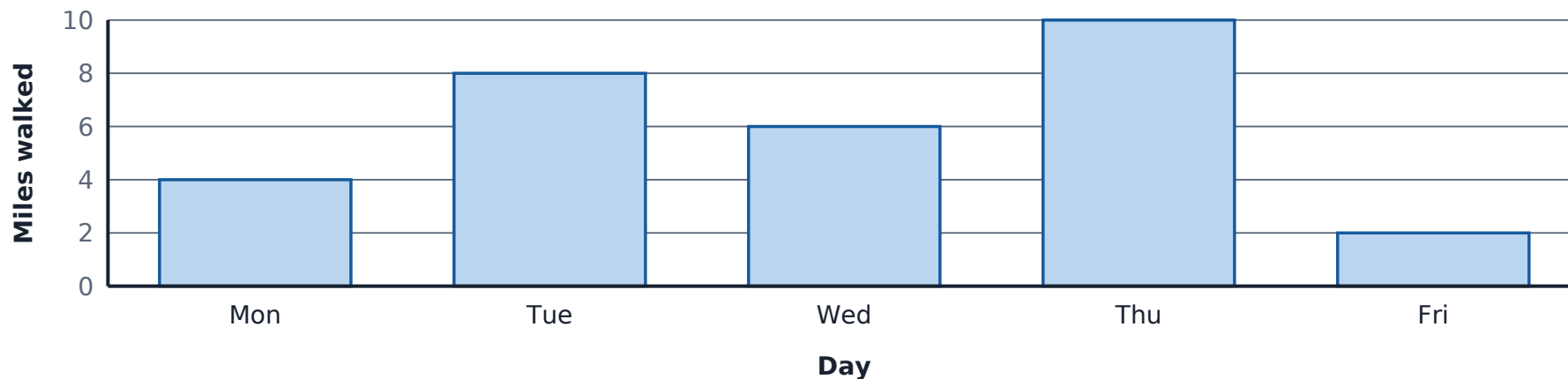
A. Thursday **B.** Friday **C.** Monday **D.** Tuesday

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 54 • BACKWARD 3 • Q3/5

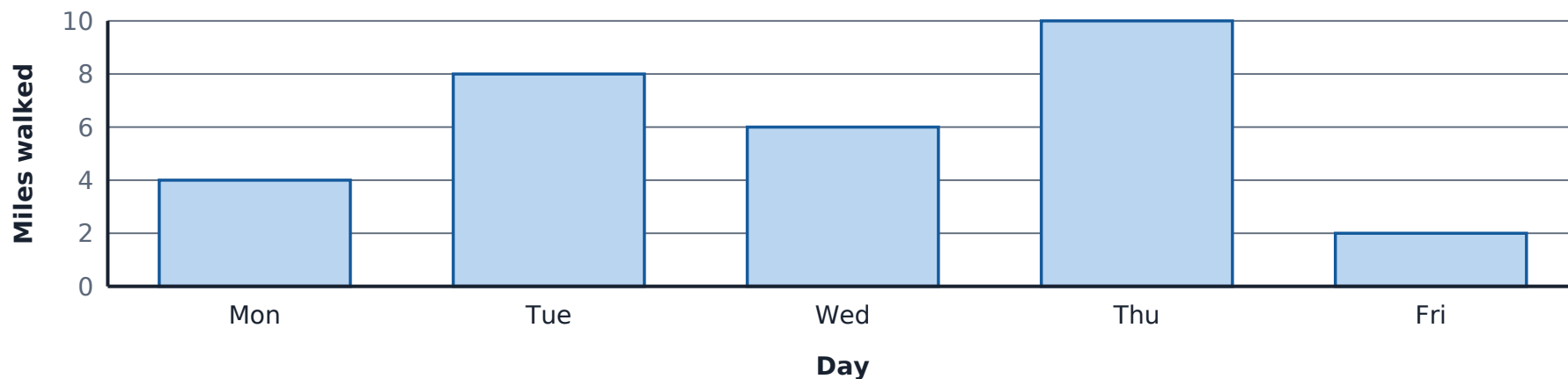
How many more miles were walked on Thursday than on Friday?



Miles: _____

True or false?

Each grid line on the scale stands for 2 miles.



A. True

B. False

SET 54 • 5.G.A.2

FORWARD 1

Find distances in all four quadrants • Grade 6 • The Number System • 6.NS.C.8

Find the distance between $(-4, 2)$ and $(5, 2)$.

Distance: _____

Find the distance between $(3, -6)$ and $(3, 1)$.

Distance: _____

On a map, the store is at $(-2, -3)$ and the bank is at $(-2, 4)$. Each unit is 1 block.

How far apart are they?

A. 7 blocks

B. 1 block

C. 4 blocks

D. 6 blocks

A rectangle has vertices at $(-3, 1)$, $(2, 1)$, $(2, -2)$, and $(-3, -2)$.

What is its perimeter?

Perimeter: _____

True or false?

The distance between $(-5, 0)$ and $(5, 0)$ is 0.

A. True

B. False

SET 54 • 5.G.A.2

FORWARD 2

Use a scale on a map grid • Grade 7 • Geometry • 7.G.A.1

On a map grid, each unit is 2 miles. Two towns are at (1, 3) and (6, 3).

What is the actual distance between them?

Distance: _____

On a floor plan, each unit is $1\frac{1}{2}$ feet. A wall goes from (0, 0) to (0, 8).

How long is the real wall?

Length: _____

A scale drawing uses 1 grid unit = 5 m. A path on the grid is 7 units long.

How long is the real path?

A. 12 m

B. 35 m

C. 2 m

D. 7 m

A garden plan has corners at $(0, 0)$, $(6, 0)$, $(6, 4)$, and $(0, 4)$. Each unit is 3 feet.

What are the real length and width?

Length and width: _____

True or false?

If each grid unit is 10 km, the points (2, 5) and (2, 9) are 4 km apart.

- A. True
- B. False

Attributes of a category belong to all of its subcategories

MAIN	Grade 5 main question	5.G.B.3
BACKWARD 1	Quadrilaterals and their attributes	3.G.A.1
BACKWARD 2	Parallel lines, perpendicular lines, and angles	4.G.A.1
BACKWARD 3	Classifying shapes by sides and angles	4.G.A.2
FORWARD 1	Draw polygons in the coordinate plane	6.G.A.3
FORWARD 2	Draw shapes with given conditions	7.G.A.2

SET 55 • 5.G.B.3

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.B.3

True or false?

Every parallelogram has opposite sides of equal length. A rhombus is a parallelogram, so every rhombus has opposite sides of equal length.

A. True

B. False

All parallelograms have two pairs of parallel sides.

Which shape must also have two pairs of parallel sides?

- A.** Kite
- B.** Pentagon
- C.** Triangle
- D.** Rhombus

A rhombus is a quadrilateral with 4 sides of equal length.

Is every square a rhombus? Explain.

Answer: _____

Explanation: _____

True or false?

All quadrilaterals have 4 right angles.

A. True

B. False

Which statement is true?

- A.** Every parallelogram is a square.
- B.** Every square is a rectangle.
- C.** Every quadrilateral is a rectangle.
- D.** Every rectangle is a square.

SET 55 • 5.G.B.3

BACKWARD 1

Quadrilaterals and their attributes • Grade 3 • Geometry • 3.G.A.1

How many sides does every quadrilateral have?

Sides: _____

True or false?

Triangles, rectangles, and squares are all quadrilaterals.

A. True

B. False

Which shape is NOT a quadrilateral?

- A.** Rectangle
- B.** Square
- C.** Rhombus
- D.** Triangle

True or false?

A square has 4 sides of equal length.

A. True

B. False

Name a quadrilateral that always has 4 right angles.

Shape: _____

SET 55 • 5.G.B.3

BACKWARD 2

Parallel lines, perpendicular lines, and angles • Grade 4 • Geometry • 4.G.A.1

True or false?

Parallel lines are always the same distance apart.

- A. True
- B. False

Which angle measure is a right angle?

- A.** 90°
- B.** 180°
- C.** 45°
- D.** 120°

What do we call two lines that meet to form right angles?

Answer: _____

True or false?

An obtuse angle is smaller than a right angle.

A. True

B. False

How many pairs of parallel sides does a rectangle have?

A. 4

B. 0

C. 2

D. 1

SET 55 • 5.G.B.3

BACKWARD 3

Classifying shapes by sides and angles • Grade 4 • Geometry • 4.G.A.2

True or false?

A shape with no parallel sides can still be a quadrilateral.

A. True

B. False

Which kind of triangle has a right angle?

- A.** An equilateral triangle
- B.** An acute triangle
- C.** A right triangle
- D.** An obtuse triangle

How many right angles does a square have?

Right angles: _____

True or false?

A rhombus always has 4 right angles.

A. True

B. False

Which shape always has two pairs of parallel sides?

A. Rectangle

B. Kite

C. Triangle

D. Pentagon

SET 55 • 5.G.B.3

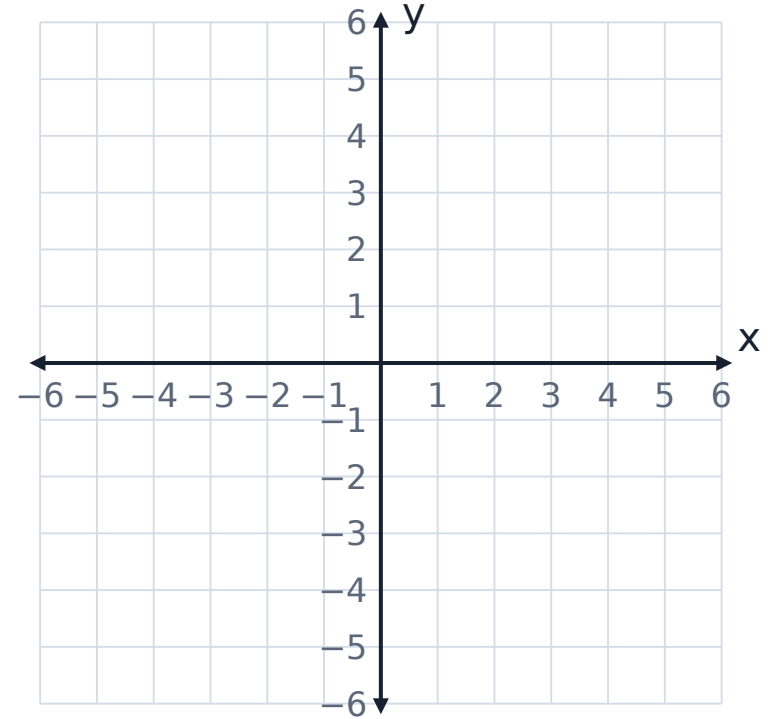
FORWARD 1

Draw polygons in the coordinate plane • Grade 6 • Geometry • 6.G.A.3

Plot $A(-2, 1)$, $B(3, 1)$, $C(3, -3)$, and $D(-2, -3)$. Connect them in order.

What is the most specific name for the polygon?

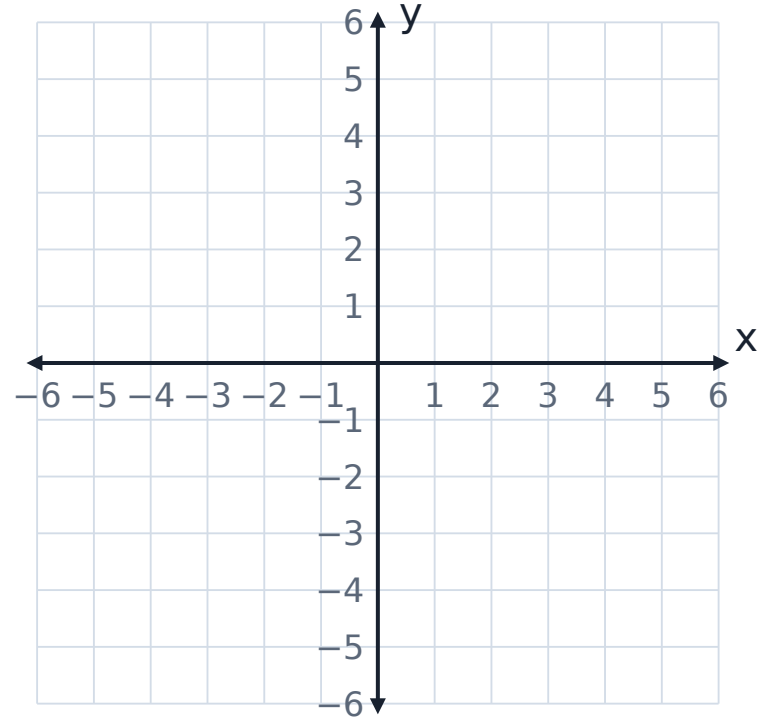
Most specific name: _____



Plot $E(0, 3)$, $F(3, 0)$, $G(0, -3)$, and $H(-3, 0)$. Connect them in order.

What is the most specific name for the polygon?

Most specific name: _____



The points $(-1, 2)$, $(4, 2)$, and $(4, -1)$ are three vertices of a rectangle.

What is the fourth vertex?

A. $(1, -1)$

B. $(-1, -1)$

C. $(-1, 4)$

D. $(4, 2)$

What is the length of the side from $(-2, 5)$ to $(4, 5)$?

Length: _____

True or false?

The points $(0, 0)$, $(4, 0)$, $(4, 2)$, and $(0, 2)$ are the vertices of a square.

- A. True
- B. False

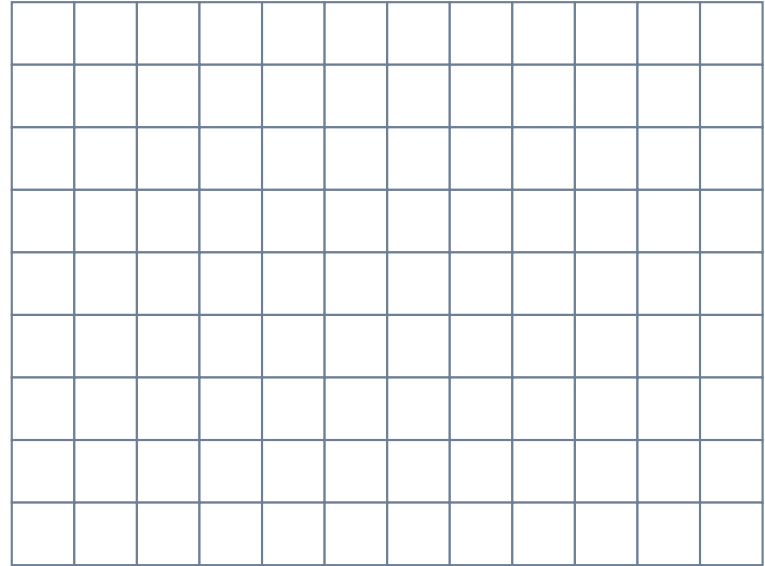
SET 55 • 5.G.B.3

FORWARD 2

Draw shapes with given conditions • Grade 7 • Geometry • 7.G.A.2

Draw a triangle with sides of 3 units, 4 units, and 5 units. Put the 3-unit and 4-unit sides along grid lines.

Classify the triangle by its angles.



Kind of triangle: _____

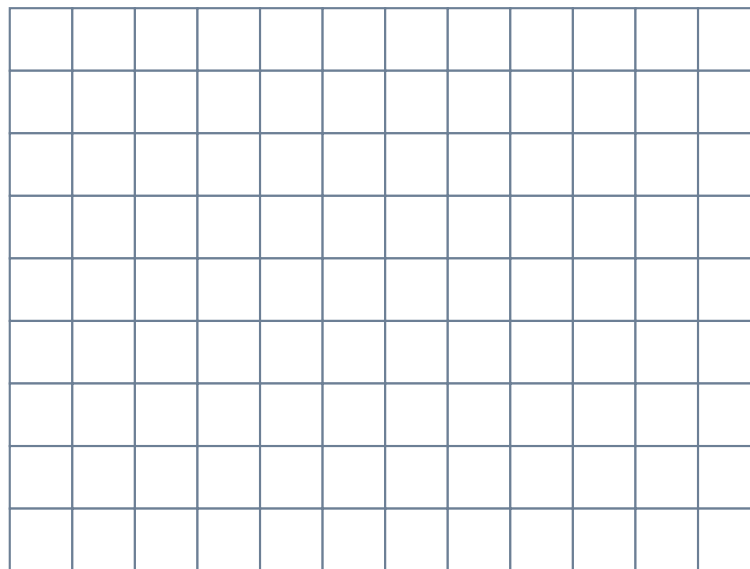
Can a triangle have sides of 2 cm, 2 cm, and 5 cm?
Explain.

Answer: _____

Explanation: _____

Draw a triangle with angles of 90° , 45° , and 45° . Make its two shorter sides 4 units long, along grid lines.

Could a triangle with these same angles be drawn in a different size?



Answer: _____

Can a triangle have angles of 100° , 50° , and 40° ?
Explain.

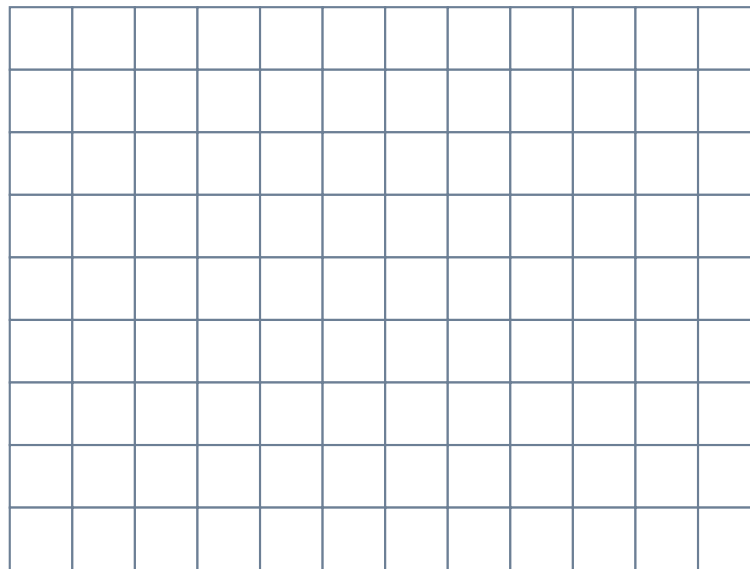
Answer: _____

Explanation: _____

Draw a quadrilateral that has exactly one pair of parallel sides.

What is the name of this kind of quadrilateral?

Name: _____



Classify quadrilaterals in a hierarchy based on their properties

MAIN	Grade 5 main question	5.G.B.4
BACKWARD 1	Quadrilaterals	3.G.A.1
BACKWARD 2	Classifying by parallel sides and right angles	4.G.A.2
FORWARD 1	Area of parallelograms, rhombuses, and trapezoids	6.G.A.1
FORWARD 2	Draw quadrilaterals and triangles with given conditions	7.G.A.2

SET 56 • 5.G.B.4

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.B.4

Which list goes from the most general category to the most specific?

- A.** Square, rectangle, parallelogram, quadrilateral
- B.** Rectangle, square, quadrilateral, parallelogram
- C.** Quadrilateral, parallelogram, rectangle, square
- D.** Parallelogram, quadrilateral, square, rectangle

True or false?

A square is a rectangle, a rhombus, and a parallelogram.

A. True

B. False

Which of these categories does every rhombus belong to?

quadrilateral, parallelogram, rectangle, square

Categories: _____

A shape has 4 sides of equal length and 4 right angles.

What is the most specific name for it?

- A.** Rectangle
- B.** Rhombus
- C.** Square
- D.** Quadrilateral

True or false?

Every parallelogram is a rhombus.

A. True

B. False

SET 56 • 5.G.B.4

BACKWARD 1

Quadrilaterals • Grade 3 • Geometry • 3.G.A.1

True or false?

A rectangle is a quadrilateral.

A. True

B. False

Which shape always has 4 sides of equal length?

- A.** Triangle
- B.** Rectangle
- C.** Rhombus
- D.** Kite

How many angles does a quadrilateral have?

Angles: _____

True or false?

Every quadrilateral is a square.

A. True

B. False

Name a shape that is a quadrilateral but is not a parallelogram.

Shape: _____

SET 56 • 5.G.B.4

BACKWARD 2

Classifying by parallel sides and right angles • Grade 4 • Geometry • 4.G.A.2

True or false?

A quadrilateral with 2 pairs of parallel sides is a parallelogram.

A. True

B. False

Which shape always has 4 right angles?

- A.** Rectangle
- B.** Parallelogram
- C.** Kite
- D.** Rhombus

How many pairs of parallel sides does a parallelogram have?

Pairs: _____

True or false?

The sides of a square that meet at a corner are perpendicular.

- A.** True
- B.** False

Which description fits every rhombus?

- A.** No parallel sides
- B.** 3 sides
- C.** 4 sides of equal length
- D.** 4 right angles

SET 56 • 5.G.B.4

FORWARD 1

Area of parallelograms, rhombuses, and trapezoids • Grade 6 • Geometry •
6.G.A.1

A parallelogram has a base of 10 cm and a height of 6 cm.

What is its area?

Area: _____

A rhombus has a base of 7 in. and a height of 5 in.
What is its area?

Area: _____

A trapezoid has bases of 4 m and 10 m and a height of 6 m. It is cut into a 4-m by 6-m rectangle and two triangles, each with a base of 3 m and a height of 6 m.

What is the area of the trapezoid?

- A.** 84 square meters
- B.** 42 square meters
- C.** 60 square meters
- D.** 24 square meters

A trapezoid has bases of 6 ft and 10 ft and a height of 4 ft.

What is its area?

Area: _____

True or false?

A parallelogram has a greater area than a rectangle with the same base and the same height.

A. True

B. False

SET 56 • 5.G.B.4

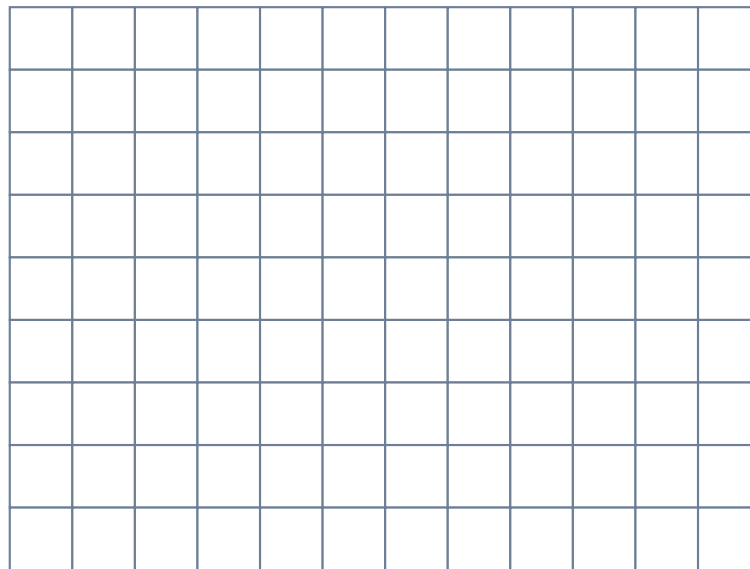
FORWARD 2

Draw quadrilaterals and triangles with given conditions • Grade 7 • Geometry •
7.G.A.2

Draw a rectangle that is not a square.

Write its side lengths.

Side lengths: _____

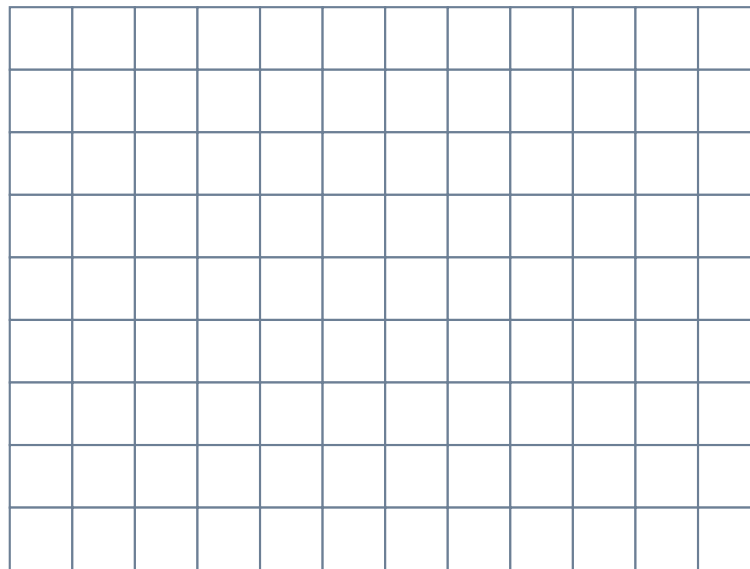


Draw a rhombus that is not a square, with every side 5 units long.

(Hint: a side can go 3 units across and 4 units up.)

How do you know it is not a square?

Reason: _____



Can a quadrilateral have exactly 3 right angles? Explain.

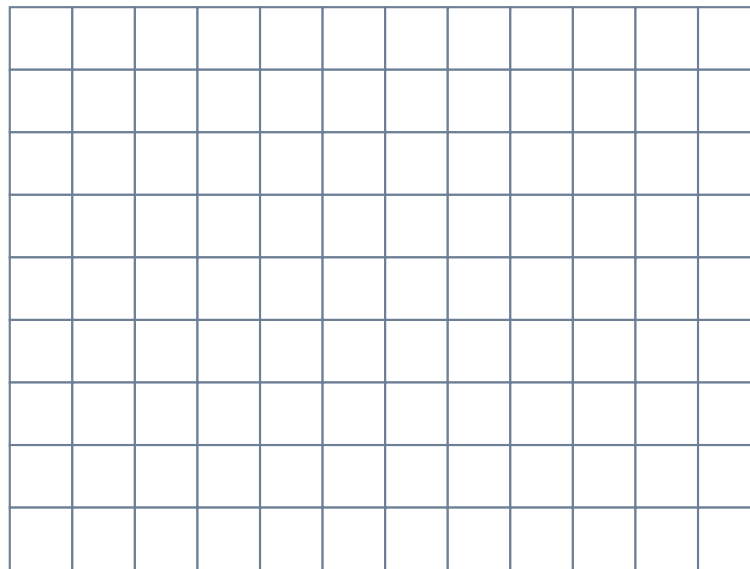
Answer: _____

Explanation: _____

Draw a parallelogram that is not a rectangle, with a base of 6 units and a height of 3 units.

What is its area?

Area: _____



A triangle has sides of 6 cm, 6 cm, and 6 cm. Can triangles of different shapes be drawn with these sides? Explain.

Answer: _____

Explanation: _____

Classify triangles by their sides and angles

MAIN	Grade 5 main question	5.G.B.4
BACKWARD 1	Right, acute, and obtuse triangles	4.G.A.2
BACKWARD 2	Kinds of angles	4.G.A.1
BACKWARD 3	Triangles	2.G.A.1
FORWARD 1	Area of triangles	6.G.A.1
FORWARD 2	Write and solve equations for unknown angles	7.G.B.5

SET 57 • 5.G.B.4

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.B.4

A triangle has sides of 6 cm, 6 cm, and 6 cm.

Which name fits it?

- A.** Obtuse
- B.** Scalene
- C.** Equilateral
- D.** Right

A triangle has angles of 90° , 50° , and 40° .
Classify it by its angles.

Classification: _____

True or false?

An isosceles triangle has at least two sides of equal length. Every equilateral triangle is also an isosceles triangle.

A. True

B. False

A triangle has sides of 3, 5, and 7 units and an angle of 120° .

Which name fits it?

A. Acute equilateral

B. Right scalene

C. Right isosceles

D. Obtuse scalene

A triangle has angles of 70° , 70° , and 40° .
Classify it by its angles and by its sides.

By angles: _____

By sides: _____

SET 57 • 5.G.B.4

BACKWARD 1

Right, acute, and obtuse triangles • Grade 4 • Geometry • 4.G.A.2

True or false?

A right triangle has exactly one right angle.

A. True

B. False

Which angle measure is obtuse?

- A.** 10°
- B.** 45°
- C.** 130°
- D.** 90°

A triangle has one angle greater than 90° .
What kind of triangle is it?

Kind: _____

True or false?

In an acute triangle, all three angles are less than 90° .

A. True

B. False

Which angle measure is acute?

- A.** 100°
- B.** 180°
- C.** 60°
- D.** 90°

SET 57 • 5.G.B.4

BACKWARD 2

Kinds of angles • Grade 4 • Geometry • 4.G.A.1

True or false?

A right angle measures 90° .

A. True

B. False

What kind of angle measures 35° ?

- A.** Right
- B.** Straight
- C.** Obtuse
- D.** Acute

What kind of angle measures 150° ?

Kind: _____

True or false?

Two right angles together make 360° .

A. True

B. False

What kind of angle measures exactly 90° ?

Kind: _____

SET 57 • 5.G.B.4

BACKWARD 3

Triangles • Grade 2 • Geometry • 2.G.A.1

How many sides does a triangle have?

Sides: _____

True or false?

A triangle has 3 angles.

A. True

B. False

Which shape has 3 sides?

- A.** Square
- B.** Hexagon
- C.** Pentagon
- D.** Triangle

True or false?

All triangles have sides of equal length.

A. True

B. False

How many vertices (corners) does a triangle have?

Vertices: _____

SET 57 • 5.G.B.4

FORWARD 1

Area of triangles • Grade 6 • Geometry • 6.G.A.1

A right triangle has legs that are 6 cm and 8 cm long.
What is its area?

Area: _____

A triangle has a base of 10 in. and a height of 7 in.
What is its area?

Area: _____

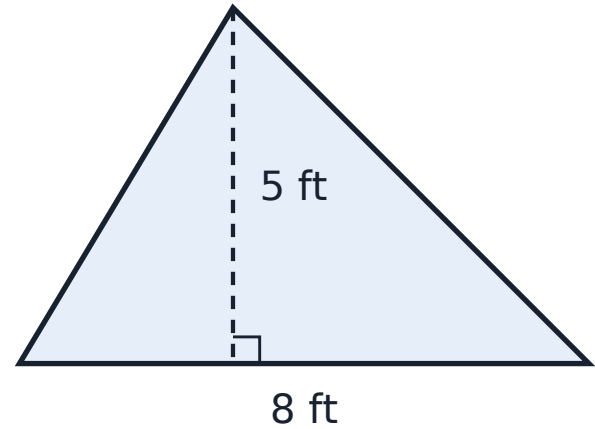
An obtuse triangle has a base of 5 m. Its height, measured outside the triangle, is 4 m.

What is its area?

- A.** 20 square meters
- B.** 40 square meters
- C.** 10 square meters
- D.** 9 square meters

Find the area of the triangle.

Area: _____



True or false?

Two triangles with the same base and the same height have the same area.

A. True

B. False

SET 57 • 5.G.B.4

FORWARD 2

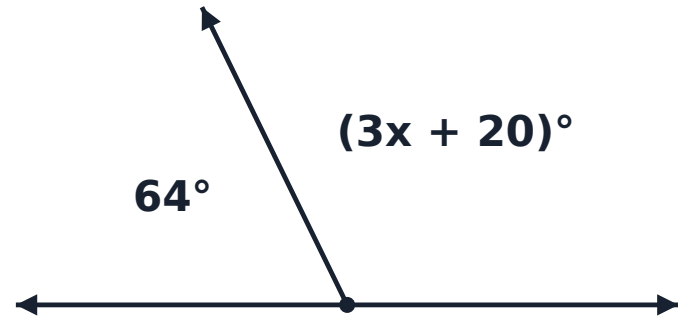
Write and solve equations for unknown angles • Grade 7 • Geometry • 7.G.B.5

The two angles form a straight line.

Write an equation and solve for x .

Equation: _____

$x =$ _____

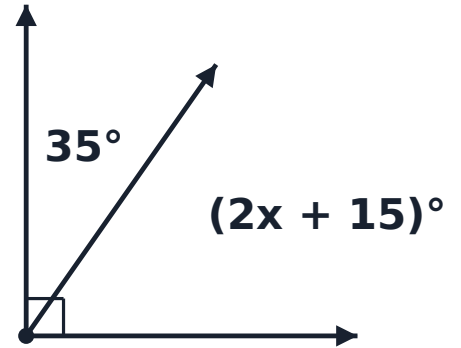


The two angles together form a right angle.

Write an equation and solve for x .

Equation: _____

$x =$ _____

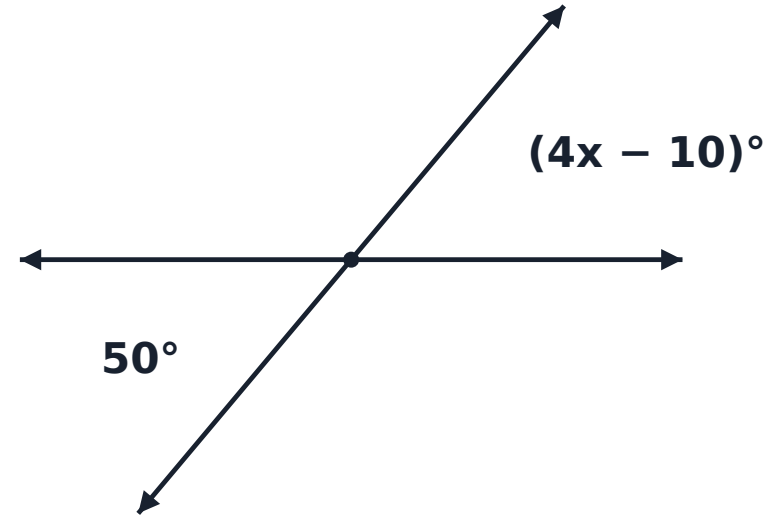


Two lines cross. The marked angles are vertical angles.

Write an equation and solve for x .

Equation: _____

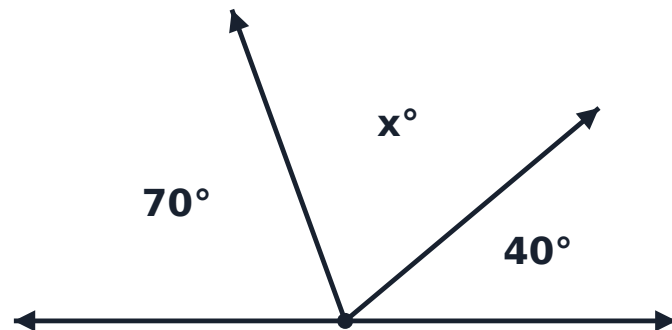
$x =$ _____



Three angles together form a straight line.

Which equation can be used to find x ?

- A.** $x = 40 \times 70$
- B.** $40 + x + 70 = 180$
- C.** $40 + x = 70$
- D.** $40 + x + 70 = 90$



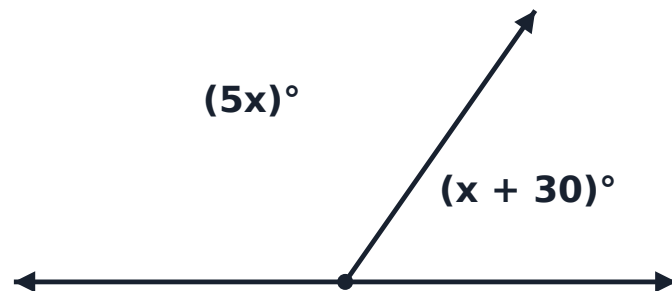
Angle ABC and angle CBD together form a straight line. Angle ABC is $(5x)^\circ$ and angle CBD is $(x + 30)^\circ$.

Write an equation, solve for x , and find both angles.

Equation: _____

$x =$ _____

Angles: _____



Place triangles in a hierarchy of categories and subcategories

MAIN	Grade 5 main question	5.G.B.4
BACKWARD 1	Classify triangles by their angles	4.G.A.2
BACKWARD 2	Shapes in different categories share attributes	3.G.A.1
FORWARD 1	Area of special triangles	6.G.A.1
FORWARD 2	Decide whether conditions give one triangle, more than one, or no...	7.G.A.2

SET 71 • 5.G.B.4

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.B.4

Which list goes from the most general category to the most specific?

- A.** Triangle, isosceles triangle, equilateral triangle
- B.** Isosceles triangle, triangle, equilateral triangle
- C.** Triangle, equilateral triangle, isosceles triangle
- D.** Equilateral triangle, isosceles triangle, triangle

True or false?

Every right isosceles triangle belongs to both the category of right triangles and the category of isosceles triangles.

- A.** True
- B.** False

Isosceles means at least two sides of equal length.

A triangle has sides of 7, 7, and 7 units. Which of these categories does it belong to?

triangle, isosceles, equilateral, scalene

Categories: _____

Isosceles means at least two sides of equal length.

Which statement is true?

- A.** Some isosceles triangles are equilateral.
- B.** No right triangle is isosceles.
- C.** Every isosceles triangle is equilateral.
- D.** Some scalene triangles are isosceles.

Explain why a triangle cannot be both a right triangle and an obtuse triangle.

Explanation: _____

SET 71 • 5.G.B.4

BACKWARD 1

Classify triangles by their angles • Grade 4 • Geometry • 4.G.A.2

True or false?

A triangle with angles of 30° , 60° , and 90° is a right triangle.

A. True

B. False

A triangle has angles of 110° , 40° , and 30° .

What kind of triangle is it?

- A.** Equilateral
- B.** Acute
- C.** Right
- D.** Obtuse

A triangle has angles of 50° , 60° , and 70° .
Is it acute, right, or obtuse?

Answer: _____

True or false?

A triangle can have two obtuse angles.

A. True

B. False

Which three angles could be the angles of an acute triangle?

- A.** 100° , 50° , 30°
- B.** 120° , 30° , 30°
- C.** 80° , 60° , 40°
- D.** 90° , 45° , 45°

SET 71 • 5.G.B.4

BACKWARD 2

Shapes in different categories share attributes • Grade 3 • Geometry • 3.G.A.1

True or false?

Rhombuses, rectangles, and squares are all quadrilaterals.

A. True

B. False

What do all quadrilaterals have in common?

- A.** 4 right angles
- B.** 4 sides
- C.** 2 pairs of parallel sides
- D.** 4 sides of equal length

Name a quadrilateral that is not a rectangle.

Shape: _____

True or false?

Every quadrilateral is a rhombus.

A. True

B. False

A square is also a ____.

- A.** rectangle
- B.** pentagon
- C.** hexagon
- D.** triangle

SET 71 • 5.G.B.4

FORWARD 1

Area of special triangles • Grade 6 • Geometry • 6.G.A.1

A right isosceles triangle has legs that are 6 cm long.
What is its area?

Area: _____

True or false?

A triangle with a base of 4 in. and a height of $3\frac{1}{2}$ in. has an area of 7 square inches.

- A.** True
- B.** False

A square with sides of 8 m is cut along a diagonal into two triangles.

What is the area of each triangle?

- A.** 64 square meters
- B.** 8 square meters
- C.** 32 square meters
- D.** 16 square meters

An isosceles triangle has a base of 10 ft and a height of 12 ft.

What is its area?

Area: _____

True or false?

A right triangle with legs of 5 units and 9 units has an area of 45 square units.

- A. True
- B. False

SET 71 • 5.G.B.4

FORWARD 2

Decide whether conditions give one triangle, more than one, or none • Grade 7
• Geometry • 7.G.A.2

How many differently shaped triangles can be drawn with sides of 5 cm, 5 cm, and 8 cm?

- A.** Exactly two
- B.** More than one
- C.** Exactly one
- D.** None

True or false?

Triangles of many different sizes can be drawn with angles of 60° , 60° , and 60° .

- A. True
- B. False

Can a triangle have sides of 3 m, 4 m, and 8 m? Explain.

Answer: _____

Explanation: _____

True or false?

A triangle can have angles of 90° , 60° , and 40° .

A. True

B. False

Can a right triangle also be isosceles? If it can, give its angle measures.

Answer: _____

Angles: _____